

South America: Updated catch reconstructions to 2018*

Emmalai Page^a, Brittany Derrick^a, Angie Coulter^a, Rachel White^b, Melanie Ang^a, Darcy Dunstan^a, Lincoln Hood^b, Veronica Relano^a, Gordon Tsui^a, Lisbeth van der Meer^c and Daniel Pauly^a

a) *Sea Around Us*, Institute for the Oceans and Fisheries, University of British Columbia, 2202 Main Mall, Vancouver, BC, V6T 1Z4, Canada

b) *Sea Around Us*- Indian Ocean, School of Biological Sciences, University of Western Australia, 35 Stirling Hwy, Crawley 6009, WA, Australia

c) Laboratório de Ecologia Pesqueira, Departamento de Engenharia de Pesca e Aquicultura, Universidade Federal de Sergipe, São Cristóvão, Brazil

d) Oceana Chile, Suecia 0155, oficina 1001, Providencia, Santiago, Chile

Abstract

Updates to earlier catch data reconstructions for the marine fisheries of South American countries (except Peru), initially covering the years 1950 to 2010 were completed to 2014 for Colombia, French Guiana and Suriname; to 2016 for Argentina, Chile (mainland, and its oceanic islands including Easter Island), Ecuador's Galapagos Islands, Guyana and Venezuela; to 2017 for Brazil mainland, Ecuador mainland and Uruguay; and to 2018 for Brazil's oceanic islands. Where applicable, these data were carried forward to 2018. The major challenge in updating these catch reconstructions was the limitations of the data reported by FAO on behalf of these countries with regard to the artisanal, subsistence and recreational sectors, and in providing the spatial resolution required to assign catches attributed to a large FAO Statistical Areas to specific Exclusive Economic Zones (EEZs). We have attempted to address these limitations by using national statistics and secondary data sources whenever possible. Details on the methods used for these updates are presented separately for each country or island (group).

Introduction

The catch of the marine fisheries of the South American countries, territories and adjacent islands (except Peru, see de la Puente *et al.* 2020) was originally reconstructed for the years 1950 to 2010. Here, we present updates to these earlier reconstructions to 2014 for Colombia, French Guiana and Suriname; to 2016 of South America for Argentina, Chile (mainland, and its oceanic islands including Easter Island), Ecuador's Galapagos Islands, Guyana and Venezuela; to 2017 for Brazil mainland, Ecuador mainland and Uruguay; and to 2018 for Brazil's oceanic islands. Where applicable, these data were carried forward to 2018.

The major challenge in updating these catch reconstructions was the limitations of the data reported by the Food and Agriculture Organization of the United Nations (FAO) on behalf of these countries with regard to the artisanal, subsistence and recreational sectors (Pauly and Charles 2015) and in providing the spatial resolution required to assign catches attributed to large FAO Statistical Areas to specific Exclusive Economic Zones (EEZs). We have addressed these limitations by using national statistics and secondary sources whenever possible. Details on the methods used for these updates are presented below for each country or island (group).

* Cite as: Page, E., B. Derrick, A. Coulter, R. White, M. Ang, D. Dunstan, L. Hood, V. Relano, G. Tsui, L. van der Meer and D. Pauly. 2020. South America: Updated catch reconstructions to 2018, p. 279-312. In: B. Derrick, M. Khalfallah, V. Relano, D. Zeller and D. Pauly (eds). *Updating to 2018 the 1950-2010 Marine Catch Reconstructions of the Sea Around Us. Part II: The Americas and Asia-Pacific*. Fisheries Centre Research Report 28(6).

Materials and Methods

Argentina

The original reconstruction of Argentina's marine fisheries catches was performed for 1950-2010 by Villasante *et al.* (2015, 2016). This account updates that reconstruction to 2016, which is then carried forward to 2018.

Reported landings data available from the FAO and the Argentine Ministry of Agriculture, Livestock, Fisheries and Food (Secretaría de Agricultura, Ganadería, Pesca y Alimentación, or SAGPyA) were compared and determined to match for 2011-2014; thus, the FAO data were used to update reported landings for 2011-2014. In the 2015 and 2016 reported data, the FAO reported data were higher than SAGPyA landings and were assumed to give a more comprehensive total. The percentages of landings originating from artisanal and industrial fisheries in 2010 were assumed to remain constant from 2010 to 2016. The 2010 ratio between unreported landings and reported landings for each sector was used to estimate unreported landings from commercial fisheries for 2011-2016.

Similarly, landings from recreational and subsistence fisheries were estimated for 2011-2016 based on the ratio between each sector and total reported landings in 2010. According to Venerus *et al.* (2017), recreational fishing regulations are becoming stricter and more widespread, resulting in the slow decline of landings. Similar trends are displayed in the subsistence fishery – the urban centers in Argentina are located away from the coastlines, and as people move away from the coast, subsistence fishing is declining (Elías *et al.* 2011). No anchor points of subsistence tonnages or rates were found. However, the calculated tonnages show a declining trend in this update.

For commercial fisheries, discards were updated for 2011-2016 based on the ratio between discards and reported landings from 2010. The taxonomic breakdown and gear breakdown of catch from each sector was maintained at the 2010 ratio for 2011-2016 to disaggregate catch by taxa.

Transition from 2016 to 2018

The catch reconstructed to 2016 was carried forward to 2018 using the semi-automated procedure outlined in Noël (2020), based on FAO landings data to 2018. Semi-automated reconstructed catch data will later be replaced by a more detailed, research-intensive update.

Marine biodiversity protection

Argentina has agreed to protect its biological diversity through the international agreements of the Convention on Biological Diversity (Aichi), United Nations Convention on the Law of the Sea, Ramsar Convention on Wetlands of International Importance, the World Heritage Convention, and Argentina is also part of the international network of UNESCO Man and the Biosphere. Argentina is a signatory to regional treaties and agreements such as the Regional Seas Convention (Marine Conservation Institute 2020).

Argentina has 71 MPAs and eight marine managed areas. Together, these areas cover 41,707 km² (Marine Conservation Institute 2020), which corresponds to 3.85% of the EEZ of 1,082,467 km² (Villasante *et al.* 2016).

The establishment of proper regulations, enforcement and education is key for better management of Argentinean MPAs. There exists, for example, large heterogeneity in the regulations used for managing marine recreational fisheries in different coastal protected areas of Argentina. This is because marine recreational fisheries regulations in Argentina are under each provincial government authority. “The management tools applied inside the Argentine coastal protected areas are the typical input (regulating fishing effort and

modalities) and output (regulating catch) controls described by the FAO (2012). One tool that could be introduced in Argentina is the engagement of local fishers as fishing guides in the recreational fishing industry, as observed in the Galapagos Marine Reserve (Ecuador). [...] So far, no monitoring programs have been put in place and the low enforcement capabilities of provincial administrations and the lack of specific funding have precluded the organization of systematic control programs for the marine environment, even in the protected areas for which specific legislation exists” (Venerus and Cedrola 2017).

One of the solutions to improve management effectiveness is applying conservation strategies from a social perspective. In the Bahía de San Antonio Protected Natural Area (BSAPNA), a framework with such a perspective was implemented to promote new strategies and improve management effectiveness. It identifies the biggest needs in order to achieve proper management, namely: delimit the autonomy of the municipal government and improve local participation for the planning and management of the reserve (use of space and resources) (Morea 2019).

Other solutions to improve conservation in Argentinean waters and ecosystems include intensive educational and advertising campaigns focused primarily on beach users and fisherman. This is based on the large amounts of debris found in the Southwestern Atlantic beaches, which mainly come from recreational and fishing activities, with a size greater than 20 mm (Becherucci *et al.* 2017).

Brazil (mainland)

The original catch reconstruction for Brazil from 1950 to 2010 was completed by Freire *et al.* (2015, 2016). In this initial reconstruction, data for each of Brazil’s 17 coastal states were reconstructed individually, then combined to complete Brazil’s mainland reconstruction from 1950 to 2010. This account documents how this initial reconstruction was updated to 2017 and then carried forward to 2018 (Figure 1).

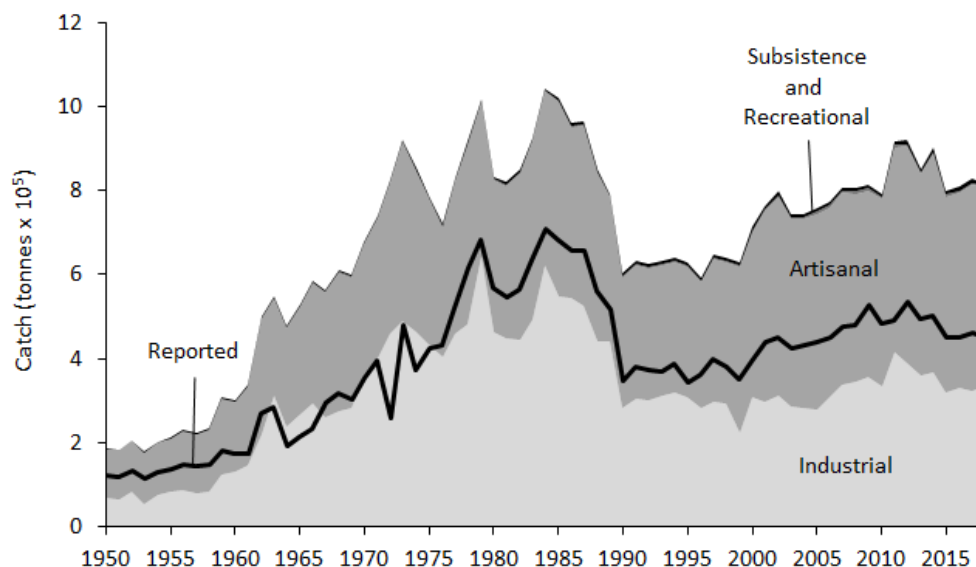


Figure 1. Reconstructed domestic catches for 1950-2018 within Brazil’s mainland Exclusive Economic Zone by fishing sector. The black overlaid line represents the official reported data.

A two-step approach for commercial catches

Brazil has not collected or published official catch data on fisheries at the national level since 2011 (Sganzerla 2017); instead, fisheries catch is reported by a few selected states and sectors. As of this writing (May 2020), the FAO has estimates of Brazilian catches to 2018, whose provenance is unknown. In the absence of other

information, and pending the completion of a thorough update by the team that completed the early reconstruction (i.e., Freire *et al.* 2015, 2016), a preliminary two-step approach was implemented here:

- i) The unpublished catch reconstruction updates (based mainly on commercial landings) performed for the States of Amapá, Ceará, Maranhão, Paraíba, Pernambuco, Piauí, Rio Grande do Sul, Rio Grande do Norte, Rio de Janeiro, São Paulo and Sergipe for 2011-2014 (Freire *et al.* unpublished data) were accepted as such;
- ii) The (per state or year) missing commercial landings to 2017 were 'filled in' by state, sector, and gear type based on the 2010 ratio and as constrained by FAO-provided data for 2017.

These two steps required numerous additional adjustments. Thus, for the States of Alagoas, Bahia, Espírito Santo, Pará, Paraná, and Santa Catarina, unreported commercial landings were reconstructed for 2011-2017 per state based on the 2010 relationship between unreported commercial landings per taxon, per sector. Also, unreported landings for Amapá, Ceará, Maranhão, Paraíba, Pernambuco, Piauí, Rio Grande do Sul, Rio Grande do Norte, and Sergipe were estimated for 2011-2014 and 2011-2015 for Rio de Janeiro and São Paulo by subtracting FAO's reported catch per taxa from the total catch reconstructed per taxon for each year and disaggregated to states and fishing sector.

When reported catches from the FAO were higher than reconstructed catches per state, no unreported catch was added. The ratio between unreported catch per taxa, per sector, per state and the reported catch for the same taxon for the last three years of catch for these states was applied to the reported landings for the years after the state-level reconstruction ended (2014 or 2015) to estimate unreported landings to 2017. Commercial gears were assigned to reported and unreported catch based on the breakdown per species from 2010.

Discards of commercial fisheries

Commercial discard amounts were corrected for each state for 1950-2010 to reflect the discard rates per gear described in the original paper (Freire *et al.* 2015). These discard rates were carried forward to 2017 and applied to commercial landings for 2011-2017.

Subsistence fishing

Subsistence catch was updated for 2011-2015 following the original methods (Freire *et al.* 2015). For 2016-2017, the total national consumption of fish and invertebrates was determined by multiplying the number of registered artisanal fishers available by year from SECAP/AVALIACAO (2019) for 2010 and the local per capita consumption rate of 14 kg per person in 2017 (Anon. 2017). We took the average from 2013 to 2015 to calculate total marine subsistence from the total estimated local consumption in those years and used this percentage to apply to total calculated consumption for 2016-2017 to estimate the subsistence marine catch. We maintained the ratio of subsistence catch per state from 2015 to 2017 to allocate total marine subsistence catch to each state. The taxonomic breakdown for 2015 was used to disaggregate subsistence catch per state for 2016-2017.

Recreational catches

Recreational catch data were updated and replaced for 1950-2010 (Freire *et al.* 2020, 2018) and updated to 2015, including improvements to estimates for Paraná and Rio de Janeiro. To estimate recreational catch for 2016-2017 in the absence of updated recreational license information, the population counts per state, per year available from Instituto Brasileiro de Geografia e Estatística (IBGE)⁷² were used based on the relationship between the population and recreational participation multiplied by an average catch rate for recreational

⁷² https://ww2.ibge.gov.br/english/estatistica/populacao/estimativa2017/estimativa_dou.shtm

fishers per state. The taxonomic breakdown for 2015 was used to disaggregate recreational catch for 2016-2017.

Transition from 2017 to 2018

This update to 2016 was carried forward to 2018 using the procedure of Noël (2020), based on FAO landings data to 2018. Semi-automated reconstructed catch data will later be replaced by a more detailed, research-intensive update. We are aware that FAO landings data are tentative, but pending a state-by-state, bottom-up catch reconstruction as earlier performed by Freire *et al.* (2015, 2016), we thought it prudent to align this update to the FAO statistics, even if their provenance cannot be established.

Marine biodiversity protection

Brazil has agreed to protect its biological diversity through the international agreements of the Convention on Biological Diversity (Aichi), United Nations Convention on the Law of the Sea, Ramsar Convention on Wetlands of International Importance, International Coral Reef Initiative, and the World Heritage Convention. Brazil is a signatory to regional treaties and agreements such as the Regional Seas Convention (Marine Conservation Institute 2020).

Brazil has 178 MPAs and seven marine managed areas. Together, these areas cover 63,630 km² (Marine Conservation Institute 2020), i.e., 2.65% of the EEZ of 2,401,000 km² (Freire *et al.* 2016), and the implemented highly protected reserves occupy 4,886 km². The use of protected areas as tools in the conservation of biodiversity is not new in Brazil but the use of protected areas in aquatic environments is more recent. Currently, there are two types of MPAs: total protected areas or no-take areas (protects ecosystems from any human interference) and sustainable use areas (permits controlled exploitation of resources) (Marine Conservation Institute 2020).

Brazilian National System for Protected Areas designates and manages these areas in Brazil. The System defines the types and categories of protected areas. The IBAMA (Instituto Brasileiro do Meio Ambiente e dos Recursos Naturais Renováveis) is the management authority in the implementation of the national environmental policy, including surveillance and control in protected areas (Giraldi-Costa *et al.* 2020). Brazil's National Commission on Biodiversity (CONABIO) approved the National Biodiversity Targets for 2020 in 2013. Brazil also prepared a Federal Government Action Plan to identify causes of biodiversity loss and enhance synergies among the ministries and other federal agencies to find adequate solutions (Marine Conservation Institute 2020).

However, effective management is still a challenge (Giraldi-Costa *et al.* 2020). In an assessment carried out in no-take and multi-use MPAs, the non-participation of local communities in the creation of the MPAs was an issue in all the studied MPAs, resulting in management challenges. "Building a systematic understanding of pre-implementation processes in other contexts can be useful, mainly because it facilitates the effective implementation of MPAs. What caught our attention was the case of the Currais NP, in which only the Federal legislature was involved in the creation of the MPA. One of the reflections made by the MPA manager of the Currais NP, was that '*It is wonderful that the National Congress wanted to create areas for the preservation of nature, but there are established ways to do this and, in this context, the National Congress skipped stages*'. Although the interviewee did not mention what skipped stages these were, we considered the attributes to be local user participation and previous studies" (Giraldi-Costa *et al.* 2020).

Some ecotourism initiatives and pilot studies have been carried out in Brazil's MPAs, such as the one in the Environmental Protection Area of Tinhare and Boipeba, which was selected due to its reefs, which provide a

marine diving trail 320 m long. This was part of the strategies of the Emancipatory Environmental Education and a product of the Community Based Marine Ecotourism (CBME). During the pilot study this activity was well accepted by local entrepreneurs, ecotourists and community, and it demonstrated economic and environmental sustainability (Rhormens *et al.* 2017).

Brazil's oceanic islands

The original fisheries reconstruction for Brazil's oceanic islands from 1950 to 2010 was completed by Divovich and Pauly (2015, 2016a, 2016b) and Pauly *et al.* (2016). This account presents an update of the reconstructions to 2018 of catches in the Exclusive Economic Zones (EEZ) of Fernando de Noronha (FN), Saint Peter and Saint Paul Archipelago (SPSPA), and Trindade and Martim Vaz Archipelago (TMV). In March 2018, marine reserves were established that partially overlapped with the Saint Peter and Saint Paul Archipelago and the Trindade and Martim Vaz Archipelago (Rodrigues 2018) bringing the total protected area to nearly 25% of Brazil's overall EEZ.

Fishing within the conservation units may only be carried out by authorized vessels that follow requirements (Rodrigues 2018). Announcement of the final design of the two large marine reserves was not without criticism by several scientists who argued that several of the most vulnerable mainland coastal areas with much higher biodiversity were excluded from the reserve design in favor of protection of open ocean areas (Fellet 2018).

Saint Peter and Saint Paul Archipelago

Reported catch data for Saint Peter and Saint Paul Archipelago were collected by national authorities for the state of Rio Grande do Norte from 1995 to 2010. However, Brazil stopped reporting national catch data in 2011 (Sganzerla 2017) and, since then, reported catch information available for select cities and states where monitoring is still required to occur.

For the Saint Peter and Saint Paul Archipelago, it was assumed that the total reported and unreported catch amounts from 2010 remained constant to 2018, assuming the fishery had not changed in recent years. Fishing was reported to continue in 2018 (even though the marine reserve took effect), as part of a partnership between the Brazilian Navy and a fishing company (Transmar - Captura, Indústria e Comércio de Pescados Ltda), wherein the company ferries 4 scientists to the island every 15 days in exchange for permission to fish within the EEZ (Fellet 2018).

To estimate the catch from this vessel, we assumed that fishing occurred year-round in 13 of the 15 days between transporting researchers to and from the mainland, and that the catch rate per day remained the same as the average catch-rate-per-vessel for 2008-2010. The percentage of reconstructed catch estimated to be reported was assumed the same as in 2010.

Reconstruction of the amount of bait and discards from fishing was carried forward for 2011-2018 using the methods described for 2010 (Divovich and Pauly 2015). The taxonomic breakdown and commercial gear breakdown for each component were held constant at the 2010 ratio for 2011-2018.

All catch from Saint Peter and Saint Paul Archipelago is considered industrial since there are no inhabitants on these islands, and industrial vessels from mainland Brazil are the only vessels fishing there.

Trindade and Martim Vaz Archipelago

Trindade Island is home to a Brazilian naval base which hosts approximately 30 soldiers stationed for four-month rotations (Anon. 2002; Santos *et al.* 2018); there are no permanent civilian inhabitants on Trindade and Martim Vaz Archipelago, and the catch here is considered entirely industrial. Catches within Trindade and Martim Vaz Archipelago were assumed to remain constant at the 2010 amount for 2011-2018.

To reconstruct catches for 2018 (the year when the marine reserve was established), the number of soldiers ($n = 32$) was multiplied by the three four month deployment periods over the year to estimate the total number of people on the island in 2018 (Santos *et al.* 2018). According to a research study by a biologist, Hudson Pinheiro, and reported by Fellet (2018), each soldier is permitted to take 15 to 100 kg of dressed fish upon leaving Trindade and Martim Vaz Archipelago. Additionally, the soldiers are reported to catch about 2.5 tonnes of parrotfish and 2.5 tonnes of groupers per year (Fellet 2018), which is considered excessive. These catches are here considered to originate from a subsistence fishery.

The landings of bait and discarded catch from longline fisheries were calculated for 2011-2018 by maintaining the methods as described for 2010 (Divovich and Pauly 2015). The catch was disaggregated by taxa and by commercial gear type using the 2010 ratios to 2018.

Fernando de Noronha

Fernando de Noronha is the only inhabited Brazilian oceanic archipelago; thus, there are artisanal and subsistence catches happening here. However, updated information on the effort and catch-per-unit-effort of the fisheries was not found for 2011-2018.

Thus, it was assumed that the small-scale catch was used to supply the seafood consumed by the population on Fernando de Noronha. Estimates of the population from Brazil's census from 1991, 2000, 2010, and 2018 (Anon. 2018) were used as anchor points and the population was interpolated between years. The reconstructed catch was divided by the population for the years 1991-2010 to estimate a per capita small-scale catch rate. The trend in the per capita catch rate was then applied to the population from 2010 to 2018.

No new information was found on the number of octopus fishers after 2010. Thus, to update the catch of this fishery, the number of traditional octopus fishers were assumed to have remained the same as in 2010 for 2011-2018, and the non-traditional octopus fishers were extrapolated for 2011-2018. The subsistence catch rate was assumed to have remained constant from 2010 to 2018, and the artisanal catch was extrapolated from its current rate of increase for 2010-2018.

To reconstruct the amount of bait caught in Fernando de Noronha from 2011 to 2018, the ratio between amount of bait and the amount of catch caught with longline and small-scale line gears from 2010 was applied to the landings from these gear types for 2011-2018.

The artisanal discard rate was slowly decreasing before 2010 and this trend was extrapolated to 2018. The discard rate was 5.35% of landings in 2010 and 5.30% in 2015. The breakdowns for taxa and commercial gears for each catch component were maintained at the 2010 ratio for 2011-2018.

Recreational catch from fishing competitions in Fernando de Noronha were reconstructed from 1996 to 2015 by Freire *et al.* (2018). To estimate unknown catch for 2002-2009, the catches were interpolated between the 2001 and the 2010 estimates. The average taxonomic of recreational fishing competitions (Freire *et al.* 2018) was held constant and applied to recreational catches for 1996-2009.

Finally, the time series ending before 2018 were carried forward to that year by extrapolating, in the absence of contrary evidence, catches based on the last five years of reconstructed catch data.

Marine biodiversity protection

Brazil has agreed to protect the biodiversity of its oceanic islands through the international agreements of the Convention on Biological Diversity (Aichi), United Nations Convention on the Law of the Sea, Ramsar Convention on Wetlands of International Importance, International Coral Reef Initiative, and the World Heritage Convention. Brazil is a signatory to regional treaties and agreements such as the Regional Seas Convention (Marine Conservation Institute 2020). However, “[there is still a] long way ahead with the objective of achieving the goals proposed at the Seventh Conference of the Parties to the Convention on Biological Diversity, of which the country is a signatory” (Schiavetti *et al.* 2013).

The first reserve in Brazilian waters was the Atol das Rocas, which was designated in 1979 and has 362 km² of no-take area. Now it is also a Ramsar site (designated in 2015), and together with the Archipelago of Fernando de Noronha, it is a World Heritage site since 2001 (Marine Conservation Institute 2020).

In the MPA located in the Archipelago of Fernando de Noronha, designated as a National Park since 1988 with 95 km² of no-take area (Marine Conservation Institute 2020), there is a conflict between shark-directed tourism and fishers who would like to access the no-take part of the MPA at least for some parts of the year. In a study that used a contingent evaluation method (i.e., willingness to pay), 67 to 71% of tourists accepted to pay a symbolic increase in the fee they already pay to either visit the island or to visit the no-take part of the MPA. The fee could be used to compensate the fishers and solve the conflict. This is an example that highlights the importance for MPAs to “outline clear compensatory mechanisms to support the transition from a resource-consumptive economy to non- or less consumptive alternatives for those directly affected by conservation, who might find it difficult to deal with short or long-term loss of access to a country's natural resources (Agardy *et al.* 2011)” (Lopes *et al.* 2018).

Chile (mainland and Juan Fernandez/Desventuradas Islands)

Since the original reconstruction from 1950 to 2010 of marine fisheries catches from mainland Chile (van der Meer *et al.* 2015, 2016) and for the Juan Fernandez/Desventuradas Islands (Zylich and van der Meer 2015, 2016), updated national and FAO data have become available which enabled an update to 2016, and a carry-forward to 2018.

These updates included a few retroactive changes made to match the most recent version of the national data (Figure 1). For example, jumbo flying squid (*Dosidicus gigas*) catches were updated back to 2001 to match the latest catch statistics. National data were made available by the Chilean fisheries organization, SERNAPESCA (SERNAPESCA 2016).

Artisanal and industrial fisheries were fully reported within the national data. Industrial catch totals were split between the Mainland Chile and ‘Outside of Exclusive Economic Zone (EEZ)’ areas with a 70%-30% ratio respectively. Artisanal unreported landings were calculated as an additional 10% of the reported landings based on the methods in the previous reconstruction (Zylich and van der Meer 2015). Industrial unreported discards were calculated as an additional 11% of the reported landings, also based on the previous reconstruction (Zylich and van der Meer 2015).

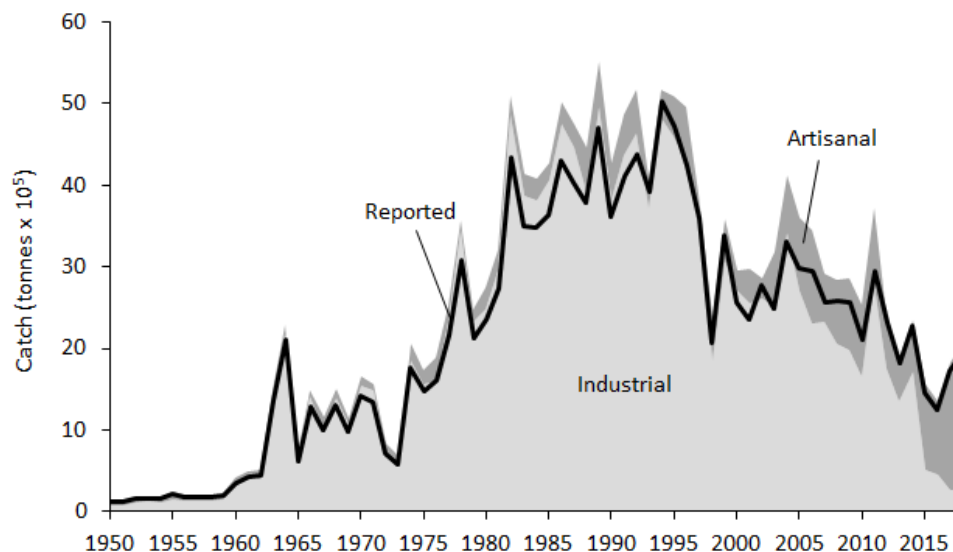


Figure 1. Reconstructed domestic catch for Chile by fishing sector for 1950-2018.

Industrial landing to mainland Chile

Industrial unreported landings are assumed to be composed entirely of unreported fish reduced to fishmeal and fish oil (FIFO). The methods used in the original reconstruction (van der Meer *et al.* 2015) were altered slightly to fit new parameters of the reported data. Reported fishmeal production was multiplied by the yield factor of 22.5% to convert to the tonnage of fish needed to produce the reported amount of fishmeal; this tonnage was compared to the sum of fish used in fishmeal production. Based on the original methods, this sum includes *Clupea bentincki*, *Engraulis ringens*, and 70% of *Trachurus murphyi* national reported totals. The difference between the reported tonnages compared to the leftover tonnages needed to produce the reported amount of fishmeal was then split into the four main species used for fishmeal: *Clupea bentincki*, *Engraulis ringens*, *Trachurus murphyi*, and *Sardinops sagax*.

The ratios used in this split were determined in the original reconstruction (van der Meer *et al.* 2015) and then split between the Mainland Chile and “Outside of EEZ” areas according to the previous ratio. In some cases, the difference between the reported fish and the fish used for fishmeal was negative – it was assumed for these cases (2012, 2014, and 2015) that there were no unreported landings because all the fish used were already reported in the landings.

Fishery for jumbo flying squid

Jumbo flying squid (*Dosidicus gigas*) were omitted in the original reconstruction (van der Meer *et al.* 2015), despite being reported nationally in very high tonnages. According to Alarcon *et al.* (2008), jumbo flying squid returned to the industrial fishing scene in 2001 due to El Niño events, and they were retroactively added for the industrial sector from 2001 onward. The artisanal reported tonnages of *Dosidicus gigas* were also incorrect. Therefore, those totals were retroactively changed from 2001 to 2018.

Marine biodiversity protection

Chile has agreed to protect its biological diversity through the international agreements of the Convention on Biological Diversity (Aichi), United Nations Convention on the Law of the Sea, Ramsar Convention on Wetlands of International Importance, the World Heritage Convention. It is also part of the international network of UNESCO Man and the Biosphere. Chile is a signatory to regional treaties and agreements such as the Regional Seas Convention (Marine Conservation Institute 2020).

Chile has 60 MPAs and seven marine managed areas. Together, these areas cover 491,951 km² (Marine Conservation Institute 2020), which equals 13.48% of the EEZ of 3,648,534 km² (van der Meer *et al.* 2016; Zylich and van der Meer *et al.* 2016; Zylich *et al.* 2016). The implemented, highly protected reserves occupy 12% of the EEZ. “Marine protected areas in Chile include federally designated marine parks, marine reserves, and coastal marine protected areas among others. MPAs include areas managed by federal, regional, and local governments as well as non- governmental organizations. [...]. The Chilean government is currently working on a law that will put protected areas under one agency. This will facilitate a more streamlined approach to creating and managing protected areas- including marine and coastal zones” (Marine Conservation Institute 2020).

According to a very recent study carried out in protected areas of Chile, only five out of a total of 20 marine protected areas (MPAs) had management plans, and no MPA had an effective plan (Petit *et al.* 2018). “The present study shows the lack of both the effectiveness of and updates to the existent management plans for the vast majority of the national territory and raise the following question: Is it sustainable to continue adding protected areas to the national system even though it is clear that the existing support is insufficient to meet the minimum requirements for full implementation? Although adding more protected areas is necessary, especially for under protected ecosystems, it is fundamental to develop effective management for fully implementation of PAs in order to meet conservation goals and targets” (Petit *et al.* 2018).

In a comparative study to assess the establishment of MPAs in Chile through bottom-up processes, the “results show that for Navidad and Punta de Lobos stakeholders, there is a positive attitude and motivation to participate in the no-take MPAs for all indicators measured. In both cases, stakeholders had positive expectations of the benefits that conservation tools like no-take MPA may bring to the community and in both cases, there were similar economic, social and environmental motivating factors. [...]. Bottom-up participatory processes to create and manage no-take marine protected areas have been proposed as a way to scale-up marine conservation and deal with the lack of support and compliance of top-down conservation approaches. However, bottom-up conservation does not always lead to positive outcomes, thus it is increasingly important to understand the conditions that determine the establishment and implementation of these initiatives” (Oyanedel *et al.* 2016).

Juan Fernandez/Desventuradas Islands

For the Juan Fernández Islands, the 2010 tonnages by taxa were carried forward to 2018 due to lack of new data. A new no-take marine reserve was implemented in the Desventuradas Islands, and therefore it is assumed that there will be no fishing in this area (Lee 2015). In 2018, the Chilean government signed into law protection for new Marine Protected Areas (MPA) that includes 101,000 square miles around the Juan Fernández Islands. The law prohibits all fishing and other resource extraction.

Chile (Easter Island)

Reconstruction of Rapa Nui's (or Easter Island's) marine fisheries was completed for 1950-2010 by Zylich *et al.* (2014, 2016). Castilla *et al.* (2014) analyzed Easter Island's traditional and artisanal fisheries for 2000-2009, and their estimates of legal landings matched the landings estimated by Zylich *et al.* (2014) over the same time period. This account documents how this initial reconstruction, which we deem reliable, was updated to 2016, then carried forward to 2018.

Baseline data

Reported landings of the fisheries of Easter Island were available for 2011-2016 from SERNAPESCA (2016) and were used to update the landings of coastal and offshore commercial fisheries; these landings were lower than in previous years (Figure 1). Species reported by SERNAPESCA from 2011 to 2015 were separated into coastal and offshore catches, and the “Miscellaneous fishes” category was divided between coastal and offshore areas based on the 2010 ratio.

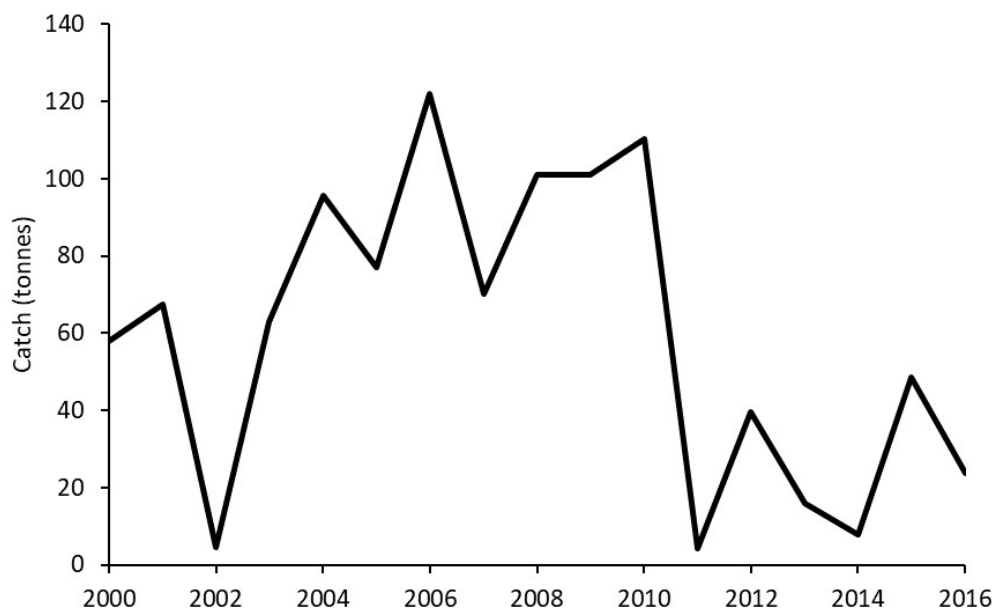


Figure 1. Reported national catch by SERNAPESCA for 2000-2016.

The fisheries resources and their protection

Castilla *et al.* (2014) described some of Easter Island’s fisheries as showing signs of overexploitation and stressed the need for improvement in collection of fishery data on Easter Island. Declines in both coastal and offshore marine resources have been described by local fishers (Aburto *et al.* 2015).

Concerns over the pressure of illegal tuna fishing and declining marine resources have led to the proposal of a 631,370 km² marine protected area (MPA) which would surround Easter Island and connect with an MPA around Sala y Gómez established in 2010 (Vaughan 2015a). Under the proposed expanded MPA, fishing by local fishers would be allowed up to 50 miles from shore on Easter Island, but no further (Vaughan 2015b). The proposal is conditional and requires the approval of Easter Island’s indigenous Rapa Nui people (Vaughan 2015b). At the time of writing, a 740,000 km² “Rapa Nui Marine Park” has been approved by referendum in 2017 and officially designated by the Chilean government as being off-limits to commercial fishing; only traditional fishing methods are allowed. Illegal landings by foreign fishing vessels were held constant at the 2010 level, as illegal fishing was not reported to have declined.

Subsistence fishing

Subsistence catches were updated for 2011-2016 using the original method of multiplying the population by per capita subsistence catch rate. A population timeline was extrapolated using the population data from Hernandez *et al.* (2000) and population anchor points provided by Mr. Rodrigo Vega (pers. comm.), based on Chilean sources. The decline in subsistence catch rate from 2002 to 2010 was assumed to continue at the same

rate for 2011-2016 in the absence of updated subsistence catch rate information. The 2010 taxonomic breakdown for subsistence catches was maintained for 2011-2016.

Transition from 2016 to 2018

The catch reconstructed to 2016 was carried forward to 2018 using the semi-automated protocol of Noël (2020), based on updated reported data from SERNAPESCA (2016). Semi-automated reconstructed catch data will later be replaced by a more detailed, research-intensive update.

Colombia (Caribbean and Pacific)

The initial reconstruction of the marine fisheries catches of Colombia, covering the years 1950 to 2005, was undertaken by Wielgus *et al.* (2007, 2010), and was updated to 2010 by Lindop *et al.* (2015) and Wielgus *et al.* (2016a, 2016b). Here, we document how these initial reconstructions were updated to 2014, then carried forward to 2018.

Baseline data

Landings from industrial shrimp fisheries reported to the FAO were assumed to be fully reported for 2011-2014. Reported shrimp landings were estimated to comprise 7.7% of the total catch associated with the shrimp fishery for 2011-2014, with the remainder of catch composed of unreported non-target landings and discards. The ratios of reported shrimp catch to non-target landings and discards from 2010 were applied to the total shrimp landings to calculate unreported catch for 2011-2014. Unreported landings and discards were disaggregated taxonomically using the 2010 ratio cited in the reconstruction (Lindop *et al.* 2015).

Landings of large pelagic taxa caught by industrial fisheries were not considered in the reconstruction as they were estimated in a separate study (Coulter *et al.* 2020). All other landings of taxa reported to the FAO were divided into the industrial and artisanal sectors for 2011-2014 based on the 2010 percentage breakdown.

Miscellaneous marine fishes were taxonomically disaggregated using the 2010 breakdown for 2011-2014.

Industrial fisheries also target anchoveta (*Cetengraulis mysticetus*) and thread herring (*Opisthonema* spp.) in the Pacific and spiny lobster (*Panulirus argus*) and queen conch (*Lobatus gigas*) in the Caribbean.

Subsistence catch

Subsistence catch were calculated, for both the Caribbean and Pacific coasts for 2011-2014 using the ratio of the 2010 reconstructed subsistence catch to reported FAO landings. The taxonomic breakdown of subsistence catches from 2010 was applied to landings for 2011-2014.

Transition from 2016 to 2018

The catches reconstructed to 2016 was carried forward to 2018 using the semi-automated procedure outlined in Noël (2020), based on FAO landings from Columbia in the Caribbean (FAO's Western Central Atlantic) and along the Pacific coast (FAO's Eastern Central Pacific) to 2018. The catch data thus updated will be later replaced by a more detailed, research-intensive update.

Marine biodiversity protection

Colombia has agreed to protect biological diversity through the international agreements of the Convention on Biological Diversity (Aichi), Ramsar Convention on Wetlands of International Importance, the International Coral Reef Initiative, and the World Heritage Convention. It is also part of the international network of UNESCO Man and the Biosphere. Colombia is a signatory to regional treaties and agreements such as the

Regional Seas Convention (Marine Conservation Institute 2020). These commitments, together with the awareness of marine biodiversity loss and fisheries decline by public and private organizations, have maintained MPAs for conservation goals (Ramirez 2016).

Colombia has 28 MPAs and six marine managed areas. Jointly these areas cover 94,218 km² (Marine Conservation Institute 2020), contributing to 12.66% of the EEZ of 744,317 km² (Wielgus *et al.* 2016a, 2016b). Some of the largest marine MPAs in Colombia are the ‘Area Marina Protegida de la Reserva de Biosfera Seaflower’ (designated in 2005 with a total area of 61,131 km² and 2,330 km² of no-take area; Digital Observatory for Protected Areas 2019; Marine Conservation Institute 2020) and the ‘Yurupari-Malpelo’ (designated in 2017 with 26,871 km²; Digital Observatory for Protected Areas 2019; Marine Conservation Institute 2020).

MPAs in Colombia still need to overcome certain challenges to reach governance effectiveness. “Barriers and opportunities for improving MPA governance are related to both government and coastal community stakeholders, and include lack of implementation of participatory policies, limited institutional and community organization capacity, loss of self-regulatory fishing practices, and violence among others. Partnerships among NGOs, private organizations, communities and government together with recent afro-descendant community organization and leadership represent key opportunities for fostering meaningful participation of communities in MPA planning/management and for improving MPA governance” (Ramirez 2016).

In a study carried out to calculate the value of the provision of fishing resources resulting from an increase of marine protected areas in Colombia, the findings reflect that “despite the reduction of the area available for fishing, in the mid-term, the protected areas generate increased levels of biomass and greater benefits associated to the fishing activity, because of the spillover effect. In that sense, the marine protected areas constitute a valid alternative for fishery conservation with the potential to generate economic benefits in the midterm” (Cuervo-Sánchez *et al.* 2018).

Ecuador (mainland)

The fisheries of Ecuador’s mainland were reconstructed for the period from 1950 to 2010 (Alava *et al.* 2015, 2016). This account documents how this initial reconstruction was updated to 2017, then carried forward to 2018.

Baseline data

Ecuador has three major industrial fisheries, for 1) small pelagic fishes, 2) large pelagic fishes and 3) shrimp fisheries.

Small and large pelagic fish

The landings of small pelagic fish were reconstructed using the ratio of FAO reported landings to estimated unreported landings from the 2010 reconstruction and applied to the reported landings to 2017. The landings are here equal to the catches, since discards are not assumed to occur.

The FAO reported landings for the large pelagic fisheries for tuna and billfish were taken as true landed catches but were excluded here as they are reported upon in a separate study (Coulter *et al.* 2020).

The shrimp fishery and its bycatch

The shrimp catches reported to the FAO were assumed to be an accurate report for 2011-2017. Non-target bycatch from shrimp fisheries was estimated with the ratio of 2.94 tonnes of non-targeted bycatch to 1 tonne of shrimp for 2011-2017. The taxonomic breakdown of non-target species was estimated in the original reconstruction for 'titi shrimp', i.e., Atlantic seabob (*Xiphopenaeus kroyeri*), and "other shrimp" and applied to the unreported catch. Of the non-targeted bycatch from shrimp fisheries, 23% was discarded.

Shark catches

Sharks are targeted by both Ecuador's industrial and artisanal fisheries (Jacquet *et al.* 2008). FAO reported catch of sharks were estimated to be 37% of total reconstructed shark landings. This relationship was applied to the reported landings to estimate the unreported landings. The tonnage of industrial shark catches estimated in the reconstruction for 2011-2017 was subtracted from the total FAO shark catch per year. The remainder of the shark catch was considered to be artisanal. The taxonomic breakdown from the 2017 FAO catch was applied to the total shark catch since it was the most comprehensive.

The total artisanal catch tonnage from 2010 was carried forward to 2014 and considered fully reported for 2016 and 2017. Artisanal catch was interpolated for 2015 between the 2014 and 2016 reconstructed landings. The FAO reported landings for artisanal taxa were subtracted from the total catch estimate to estimate unreported landings. Taxa were assigned to the total catch by applying the taxonomic breakdown from the reconstruction.

A new jellyfish fishery

Cannonball jellyfish (Stomolophidae) were reported to the FAO by Ecuador for the first time in 2014. This catch was assumed to be correct because fisheries for cannonball jellyfish are developing in the FAO area Eastern Central Pacific, and other countries in the region report a similar growing trend (Brotz 2017).

Subsistence fishing

Subsistence fisheries consisted of catches of cockles (*Anadara* spp.) and mangrove red crab (*Ucides occidentalis*). Since the original reconstruction estimated very little Illegal, Unreported and Unregulated (IUU) catch of *Anadara* cockles, it was assumed that the catch was equal to the FAO reported for cockles. Reported catches of *Anadara* cockles were assigned to 20% subsistence catch, with the remaining catch assigned to the artisanal sector.

Mangrove red crab subsistence catch was calculated as a per capita catch rate for people living in rural areas within 10 km of the coast. Because the coastal population data used in the original reconstruction was not current to 2017, a ratio of rural coastal peoples to total population was calculated for 2010 and applied to total population data for 2011-2017 from the World Bank (2018). The 2010 per capita subsistence catch rate of mangrove red crab was held constant for 2011-2017 and applied to population for 2011-2017.

Transition from 2017 to 2018

The catches reconstructed to 2017 for Ecuadorian mainland were carried forward to 2018 using the semi-automated procedure outlined in Noël (2020), based on FAO landings data. Semi-automated reconstructed catch data thus obtained will be later replaced by a more detailed, research-intensive update.

Marine biodiversity protection

Ecuador has agreed to protect biological diversity through the international agreements of the Convention on Biological Diversity (Aichi) and the Ramsar Convention on Wetlands of International Importance. It is also part of the international network of UNESCO Man and the Biosphere (Marine Conservation Institute 2020).

Ecuador has 19 MPAs (without considering the Galapagos Marine Reserve) and 9 marine managed areas within the EEZ of the Ecuadorian mainland, which covers 236, 556 km² (Alava *et al.* 2016).

“Ecuador’s National System of Protected Areas (NSPA) is composed of four subsystems, Natural Heritage Areas (NHA), Protected Areas of Local Governments, Community Areas, and Private Areas. The Ministry of the Environment (MoE) is responsible for crafting regulations and coordinating the management of NSPA. [...] To date, all the marine and coastal protected areas fall within the NHA subsystem, with the exception of a large number of small mangrove areas that can be classed within the Community Areas’ subsystem. [...] Some ecological benefits are also emerging, with local proposals for No Take Zones and an incentive-based scheme to restore lobster populations, which is already yielding encouraging results” (Gravez *et al.* 2013).

The Ministry of Tourism (MINTUR) and the Ministry of Environment (MAE) of Ecuador have implemented projects to sustainably use marine resources. The MAE manages the National System of Protected Areas (NSPA), which allows free access to tourists and ensures the preservation of ecosystems and services of environmental goods. In a tourist evaluation carried out in Villamil Beach National Recreation Area (VBNRA, designated in 2011 with 24 km²) and also ruled by MAE, the results suggested that visitors are responsive to the variations of travel costs. Residents are more sensitive to marginal increments in costs than tourists are (Zambrano-Monserrate *et al.* 2018). “Ecotourism is beneficial for a country like Ecuador because it is a possibility to increase the attraction of protected areas, because it provides an economic reason for protecting them, besides it provides to local population economic incomes by improving their level of life and encouraging them to respect and conserve protected areas” (Quezada Sarmiento *et al.* 2018).

Ecuador (Galápagos Islands)

The original reconstruction of the fisheries of the Galápagos Islands was completed for 1950 to 2010 (Schiller *et al.* 2013, 2015, 2016). Here, we document how this initial reconstruction was updated to 2016, then carried forward to 2018.

Baseline data

The two invertebrate fisheries previously reported by FAO for green spiny lobster (*Panulirus gracilis*) and brown sea cucumbers (*Isostichopus fuscus*) are no longer reported upon. However, although landings of green spiny lobster have not been reported since 2006, the fishery remains active (Ramírez *et al.* 2013); on the other hand, the sea cucumber fishery was closed in 2006, 2009 and 2010 (Reyes *et al.* 2013) and landings have not been reported to the FAO since 2008. Landings from the industrial tuna and billfish fisheries are reconstructed in a separate project (Coulter *et al.* 2020), and therefore, they are not addressed in this update.

Since 2004, small-scale fisheries have been stable at approximately 5000 tonnes. Because of this stability and the lack of recent fisheries information, the 2010 reconstructed values have been extended to 2016 for small-scale fisheries. This will be reassessed when current information becomes available.

Fisheries impacts

Industrial catch of sharks was held constant at the 2010 level to 2014; shark catches declined to 12.5 tonnes per year for 2015-2016, possibly as a result of overfishing. However, some of this decline, at least for 2016,

may be a result of the establishment of a new no-take reserve around the northern Galápagos Islands of Darwin and Wolf in 2016.

The effects of artisanal longline fishing in the Galápagos were investigated by Cerutti-Pereyra *et al.* (2010), who recommended that non-selective fishing gears should be avoided to protect sensitive species and the integrity of the Galapagos marine ecosystem. Eddy *et al.* (2019) reconstructed historical unfished stock biomass of heavily exploited sailfin grouper (*Mycterooperca olfax*), green spiny lobster and pronghorn lobster (*Panulirus penicillatus*) and sea cucumber based on fishers' knowledge and suggested that fishing had a higher impact on the role of changing the sailfin grouper within the ecosystem than El Niño events.

Transition from 2016 to 2018

The catches reconstructed to 2016 for the Galápagos were carried forward to 2018 using the semi-automated procedure outlined in Noël (2020), based on updated national data. Semi-automated reconstructed catch data will be later replaced by a more detailed, research-intensive update.

Marine biodiversity protection

The Galapagos Archipelago islands are surrounded by the Galápagos Marine Reserve (GMR), created in 1998 and covering an area of 138,000 km² (Moity 2018), which represents 16.52% of the Galápagos Islands EEZ of 835,541 km² (Schiller *et al.* 2016). However, the no-take area is 1,323 km², which implies that less than 1% of the GMR is protected from fishing. The GMR was created on the basis of a 40 nautical mile buffer from the smallest polygon that encompasses all islands boundaries (Moity 2018). The dimensions of the zones range from small offshore islets to longer stretches of the coastline of larger islands (Castrejón and Charles 2020). Due to its outstanding universal value, the GMR became a Natural World Heritage Site in 2001 (Moity 2018).

The GMR is open to artisanal fishing, which occurs mostly within the proximity of the coastline and the shallow seamounts (called “bajos”); the longest fishing trips are between islands to reach to the fishing grounds or to more distant bajos (Moity 2018).

The Ecuadorian government encountered difficulties to manage protected areas due to deficiencies in funding, technical capacity and infrastructure. This is why the Galápagos National Park was only a legal framework for eight years. “For this reason, it entrusted the CDF (Charles Darwin Foundation for the Galápagos Islands) with the execution of Galápagos' biodiversity inventory and conservation activities. [Formally], this situation changed through the creation of the Galápagos National Park Service (GNPS) in 1968, which received full responsibility to manage the park [on paper]” (Denkinger *et al.* 2014). However, collaboration between the CDF and GNPS continued afterwards.

“The history of conservation management in the Galápagos is complex, having been shaped by the interaction of actors with different objectives, interests, and levels of power” (Birbano *et al.* 2020). “Thus, the establishment of MPAs are also driven by external factors, as in the case of the Galápagos Marine Reserve (Barragan-Paladines and Chuenpagdee 2017). Its establishment was driven by complex geopolitical issues (officially exerted by the Ecuadorian government in relation to maintaining control of the archipelago), economic (due to the high profitability of local resources), social, and environmental (due to the health of marine resources and ecosystems in the area, due to heavy exploitation). The making of the Galápagos Marine Reserve was also influenced by the interest in promoting “nature-based” tourism or ecotourism such as shark watching and scuba diving that has become a popular activity, placing Galápagos Marine Reserve within the top ten diving sites in the world. The idea of an MPA in Galápagos was not initialized or effectively communicated within the local social-cultural context. Instead, it was heavily influenced by external

geopolitical and tourism-based economic interests, that were masked under environmental discourses (Barragan-Paladines and Chuenpagdee 2017)” (Giraldi-Costa *et al.* 2020).

French Guiana

Reconstruction of marine fisheries catches in French Guiana for 1950-2010 was performed by Harper *et al.* (2015, 2016). This account documents how this initial reconstruction was updated to 2014, then carried forward to 2018.

Baseline data

FAO landings of shrimp (*Penaeus shrimps nei*, ‘Southern brown shrimp’, or ‘Red spotted shrimp’) in 2011-2014 for French Guiana were assigned to foreign fishing by France as detailed in the original reconstruction. The ratio of discards from shrimp landings by France was maintained throughout the time series. The taxonomic ratio of discards from the shrimp fishery were also held constant at the original ratio for 2011-2014. Unreported landings of snappers in French Guiana by Venezuela were updated for 2011-2014 by holding constant the 1999 tonnage.

Artisanal landings included all reported FAO taxa but did not include ‘Yellowfin river pellona’, ‘*Penaeus shrimps nei*’, ‘Southern brown shrimp’, or ‘Red spotted shrimp’ assigned to foreign fishing entities as described above.

Subsistence catches were calculated for 2011-2014 based on the ratio between subsistence and artisanal landings for 2010. The taxonomic breakdown was maintained throughout the time period.

Cissé *et al.* (2014) evaluated the sustainability of small-scale coastal fisheries in French Guiana and concluded that improvements to the fishery are required, including reducing the level of discards and providing better monitoring.

Transition from 2014 to 2018

The catches reconstructed to 2014 for Ecuadorian mainland were carried forward to 2018 using the semi-automated procedure outlined in Noël (2020), based on FAO landings data. Semi-automated reconstructed catch data will be later replaced by a more detailed, research-intensive update.

Marine biodiversity protection

France has agreed to protect the biological diversity of French Guiana through the international agreements of the Convention on Biological Diversity (Aichi) (Marine Conservation Institute 2020).

French Guiana has three marine protected areas: the Grand Connétable, the Kaw-Roura, and the Amana Nature Reserves (Geographic Guianas 2020). Jointly, these areas cover 1,365 km² (UNEP-WCMC and IUCN 2020), which corresponds to 1% of the EEZ of 134,656 km² (Harper *et al.* 2016).

In the Kaw-Roura Nature Reserve, characterized by mangrove ecosystems, the inhabitants are very isolated and mainly devoted to hunting and fishing in the Kaw River. Due to the protection measures established after the reserve designation, they were afraid to lose their primary source of protein. The descendants of enslaved Africans, this group feared losing access to these resources due to their lack of property rights, unlike the Amerindians of Guiana, who are recognized as autochthonous (Cormier-Salem 2017).

“The implementation of the Natural Reserve of the Kaw-Roura wetland of a surface of about 98500 ha and its ecotourist valuation arouse strong tensions. Classified as a Ramsar site in November, 1993, the Natural Reserve was created by decree in March, 1998. Thanks to its vast surface, the mosaic of ecosystems and landscape (from mangrove forest to freshwater forests or “pinotières”) and the wealth of the fauna (black caiman (*Melanosuchus niger*), Matamata tortoise (*Chelus fimbriatus*), manatee (*Trichechus manatus*), agami heron (*Agamia agami*), crested Hoazin (*Opisthocomus hoazin*), red Ibis (*Eudocimus ruber*), Kaw Roura is the biggest wetland of France (1,370 km²) and has a status of Natural Sanctuary” (Cormier-Salem 2017).

In the French Guiana’s EEZ, activities such as fisheries, shipping and gas prospection are increasing. This raises concerns about potential threats to the unique ecosystems and marine biodiversity of these waters (Martinez *et al.* 2019).

In the Grand Connétable waters, seabirds and humpback whales are some of the most commonly observed animals (Martinez *et al.* 2019). The presence of other marine mammals, such as the endangered Antillean manatee (*Trichechus manatus manatus*), has been confirmed in all the rivers and estuaries of French Guinea (Castelblanco-Martinez *et al.* 2018). Other studies have shown the behavioral changes of gravid olive ridley sea turtles (*Lepidochelys olivacea*) due to the unusual estuarine conditions of the French Guiana continental shelf. These turtles are heavily affected by the bycatch of illegal drifting nets in the coastal area of the Guiana shield (Chanbault 2017).

Guyana

A reconstruction of Guyana’s marine fisheries catches was performed for 1950-2010 by MacDonald *et al.* (2015, 2016). Here, we document how this initial reconstruction was updated to 2016, then carried forward to 2018.

Baseline data

Reported landings of shrimp (mainly *Farfantepenaeus brasiliensis*, *F. notialis* and *F. subtilis*), sea bob (*Xiphopenaeus kroyeri*), finfish caught by the industrial fishery, and the catch of groups caught by the artisanal fisheries were updated to 2016 based on the FAO dataset for that year. FAO data for ‘Marine fishes nei’ was allocated to artisanal small-scale and industrial bottom trawl gears using the same ratio as in 2010. The taxonomic breakdown of ‘Marine fishes nei’ for both sectors was held constant throughout the time series. Unreported landings from artisanal fisheries were updated for 2011-2016 as determined in the original reconstruction. Discards from reported landings of sea bob and finfish taxa were calculated as a percentage of reported landings for 2011-2016 as stated in the original methods.

Subsistence fisheries

Estimates of the 2002 and 2012 population of Guyana were available from the Guyana Bureau of Statistics and used to interpolate total population between 2003 and 2011. The total population of Guyana for 2013-2014 was extrapolated from the 2012 anchor point using the interpolated rate of change. For 2015-2016, the Guyanese population was found using the World Bank database. Total subsistence catch was equal to population data multiplied by the per capita subsistence catch rate from 2000 which was held constant for 2000-2016. The taxonomic breakdown of subsistence catches was held constant throughout the time series.

Illegal fishing

Guyana recently signed the legally binding UN Agreement on Port State Measures to Prevent, Deter and Eliminate Illegal, Unreported and Unregulated Fishing (FAO 2016) which came into effect in June 2016. The international accord is intended to stop illegal fishing by requiring vessels to request permission to access

ports and to provide information to ports on their catch and activities (Anon. 2016b). Most recently Guyana had heightened concerns over illegal activity after pirates boarded a vessel, left one person dead and left three people missing (Xinhua 2016). The incident has reignited the debate over Guyana's Public Security Minister's proposal to allow fishers to carry firearms while at sea (Anon. 2015, 2016a).

Transition from 2016 to 2018

The catches reconstructed to 2016 for the Guyana were carried forward to 2018 using the semi-automated procedure outlined in Noël (2020), based on FAO landing data. Semi-automated reconstructed catch data will be later replaced by a more detailed, research-intensive update.

Marine biodiversity protection

Guyana has agreed to protect its biological diversity through the international agreements of the Convention on Biological Diversity (Aichi) (Marine Conservation Institute 2020).

Guyana has only one MPA, i.e., the managed resource use area of 'Shell Beach', designated in 2005 with a total area of 6,250 km² and 1,7 km² of reported marine area (Marine Conservation Institute 2020), corresponding to 0.0012% of an EEZ of 140,369 km² (MacDonald *et al.* 2016).

The Protected Areas Act, published in 2011, has several objectives, the main ones being the conservation of biological diversity, safeguarding ecosystem services, encouraging stewardship of Guyana's natural heritage, assisting in safeguarding Guyana's sovereignty over its natural heritage, regulating access to the nation's biological resources and establishing a protected areas commission and a protected areas trust to manage and finance the national protected areas system (Marine Conservation Institute 2020).

"The recent large-scale offshore oil discoveries in Guyana hold the promise of lucrative economic opportunities for the country and its citizens, including its indigenous population. [...] While much scholarly and political effort is turned to the identification of good regulatory and fiscal governance of the newly discovered resources, little attention has been granted to safeguarding the rights of indigenous peoples in Guyana in light of these new extractive activities" (Shapovalova 2020).

The arrival of the oil and gas sector in Guyana brought with it great environmental challenges that jeopardize Guyana's ability to meet its obligations under the Convention on Biological Diversity. It will therefore be necessary to find the right balance between the offshore oil and gas development and environmental protection (Elias-Roberts 2020). The history of the gold mining industry, currently the largest contributor to Guyana's GNP and for which the implementation of safety and environmental standards was never a priority, should not repeat itself (Howard *et al.* 2011). "Guyana has incorporated sustainable development into several acts and policy documents related to the environment. However, it is time for the government to consider incorporating sustainable development into laws aimed at specifically regulating the newly emerging petroleum sector. Guyana has the tendency to enact environmental legislation in an *ad hoc* manner, which is usually in response to specific crisis. [...]. As a result, the laws regulating marine biodiversity and forest governance have been subjected to this pattern" (Elias-Roberts 2020).

Suriname

The initial reconstruction of the catch of Suriname's marine fisheries was completed for the years 1950 to 2010 by Hornby *et al.* (2015, 2016). Here, we document how this reconstruction was updated to 2014, then carried forward to 2018.

Large pelagic fishes

In 2011, FAO began reporting Surinamese landings of large pelagic fishes such as wahoo (*Acanthocybium solandri*), common dolphinfish (*Coryphaena hippurus*), and several shark taxa. These landings were assumed to be retained bycatch from industrial tuna fisheries, which were also reported by the FAO starting in 2011. However, Surinamese landings and associated bycatch from industrial tuna fisheries are addressed separately, based on ICAAT as described by Coulter *et al.* (2020).

Industrial shrimp fishery

Retroactive changes were made to reported landings from industrial shrimp fisheries throughout the time series to match the latest FAO data version. Landings of Atlantic seabob (*Xiphopenaeus kroyeri*) and penaeid shrimp were assumed to be fully reported in 2011-2014. Discards linked to shrimp landings were assumed to continue being generated at the rate determined by the original reconstruction. Discarded bycatch associated with the seabob fishery were estimated as 12% of total trawl catch for 2011-2014. Landed bycatch was calculated as 27% of the seabob landings value for 2011-2014 and was assumed to be accounted for in reported landings of “Marine fishes nei”.

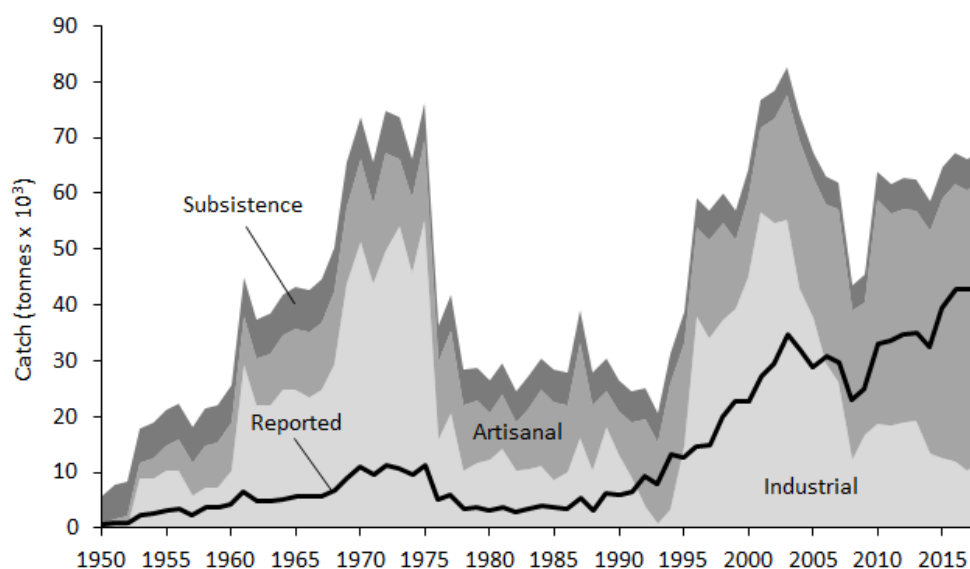


Figure 1. Reconstructed domestic catch within Suriname’s EEZ by fishing sector (1950-2018).

Demersal fish catches

The remainder of “Marine fishes nei” landings were allocated to the artisanal sector and taxonomically disaggregated using the 2010 breakdown for 2011-2014. Artisanal landings for 1950-1969 were corrected due to a calculation error in the original reconstruction. Unreported artisanal landings were maintained at 41% of total artisanal landings for 2011-2014. Landings by the fleets of Japan, South Korea and Venezuela were carried forward unaltered to 2014 at the 2010 amounts.

Subsistence catches

Subsistence catches were updated for 2011-2014 using the methods in the original reconstruction. The per capita fish consumption rates of 18 kg·person⁻¹·year⁻¹ and 9 kg·person⁻¹·year⁻¹ were applied to the rural and urban coastal populations of Suriname, respectively. Subsistence catches were disaggregated for 2011-2014 using the 2010 taxonomic breakdown.

Transition from 2014 to 2018

The catches reconstructed to 2014 for Suriname were carried forward to 2018 (Figure 1) using the semi-automated procedure outlined in Noël (2020), based on FAO landing data. Semi-automated reconstructed catch data will be later replaced by a more detailed, research-intensive update.

Marine biodiversity protection

Suriname has agreed to protect its biological diversity through the international agreements of the Convention on Biological Diversity (Aichi) (Marine Conservation Institute 2020).

Suriname has eight MPAs and no marine managed areas. Together, these areas cover 2,082 km² (Marine Conservation Institute 2020), which is 1.63% of its EEZ of 128,000 km² (Hornby *et al.* 2016). Among the four Nature Reserves, the one with the largest reported marine area is the ‘Coppename Monding’, designated in 1966 with a total area of 351 km² and a marine area of 231 km² (Marine Conservation Institute 2020).

One of the biggest multiple use management areas (MUMA) in Suriname is the Bigi Pan, which was designated in 1987 with a total area of 1537 km² and a marine area of 890 km² (Marine Conservation Institute 2020). “[The MUMA] is characterized by wetlands with mangrove forests, contains high biodiversity, and is of socio-economic, ecological and ornithological importance. However, the MUMA is overexploited and subject to competition between various income generating activities, including uncontrolled fisheries and unregulated tourism combined. Insufficient capacity of government agencies for enforcement and policy implementation and lack of communication between relevant government agencies has further contributed to unsustainable practices that diverge from ‘wise use’ and conservation. [...] Shared understanding of the need to protect natural resources in the Bigi Pan MUMA among stakeholders, the need for law enforcement to address hunting activities, and the need for increased management and monitoring capacity may bring about a collaborative process” (Djosetro and Behagel 2020).

On the basis of this research, it was concluded that protection of valuable ecosystems in Suriname needs better communication between the different stakeholders and investment in human and financial capacity of government agencies to monitor and minimize anthropogenic pressures that compromise the performance of the MUMA (Djosetro and Behagel 2020).

Uruguay

The original reconstruction of Uruguay’s marine fisheries catches was completed for 1950-2010 by Lorenzo *et al.* (2015, 2016). This account documents how this initial reconstruction was updated to 2017, then carried forward to 2018.

Baseline data

Nationally reported landings were available by sector from 2008 to 2015 (DINARA 2015) and were used to update the total reported landings from industrial fisheries for these years. In the absence of accessible nationally reported data, the ratio of FAO landings assigned to industrial fisheries in 2015 was used to disaggregate FAO landings into industrial and artisanal sectors for 2016-2017.

A portion of artisanal landings was reported by DINARA (2015) since the mid-2000s and are available for 2008-2015. Reported artisanal landings increased from 3239 tonnes in 2010 to 9397 tonnes in 2014, while the number of artisanal vessels declined from 752 vessels in 2010 to 585 vessels in 2014 (DINARA 2014b, 2015). Therefore, it was assumed that the increase in reported artisanal landings is most likely due to improvement in catch reporting for small-scale fisheries rather than a real increase in catches.

In the original reconstruction, the artisanal catch had been entirely unreported. However, this catch has since been disaggregated from 2008 onward into three categories: landings reported to the FAO, additional national reported landings that did not appear to transfer to the FAO data and unreported catch reconstructed as a ratio of reported landings. Therefore, retroactive changes were made to the 2008- 2010 fraction of the artisanal landings that were reported, such that they now include the nationally reported catch.

Recreational and subsistence fishing

Updated population information from World Bank was used to calculate the number of recreational fishers based on the 2010 percentage of the total population engaged in recreational fishing. The 2010 recreational catch rate was maintained at the 2010 level in order to calculate recreational landings for 2011-2017. The taxonomic breakdown of recreational landings was assumed to remain the same as in 2010.

Subsistence catches were estimated to be 5% of artisanal landings for 2011-2017, as described in the original report. Subsistence and artisanal landings were disaggregated by taxa for 2011-2017 using the 2010 taxonomic breakdown for each sector. Fishing gears were assigned to commercial fisheries per taxa for 2015-2017 using the percentage breakdown from 2014.

Improvement to the reporting system

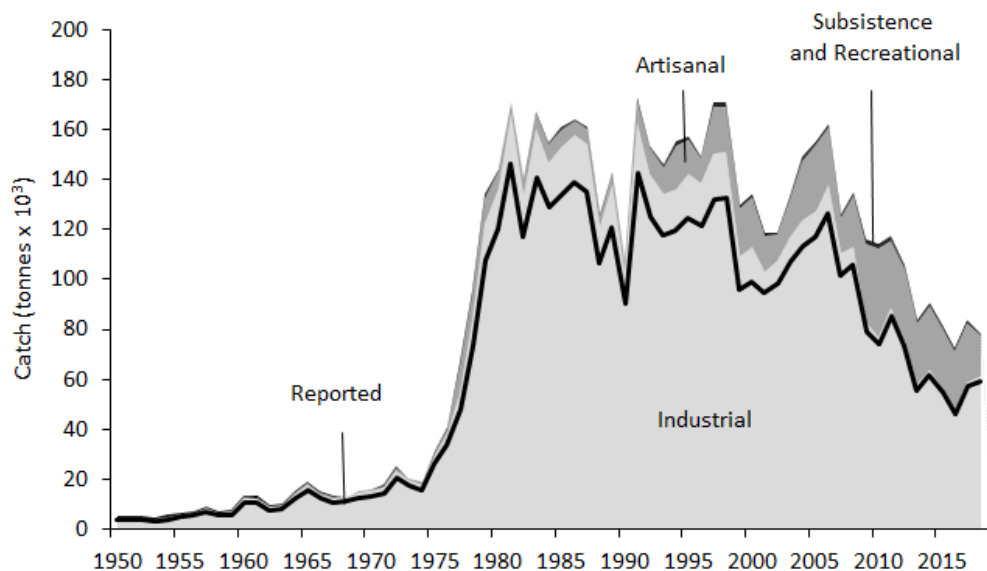
There have been improvements in information available about Uruguay's artisanal fishing sector, including general taxonomic breakdown, gear types used, investigations into catch-per-unit-effort, and spatial variability of the artisanal fleet (Defeo *et al.* 2009; Ligrone *et al.* 2014). There has also been a push for an 'Ecosystem Approach to Fisheries', introduced by the FAO and endorsed by Uruguay (Gianelli *et al.* 2018).

However, there is still no evidence for the level of underestimation (unreported landings) of catch by official sources for 2011-2017. Therefore, the total artisanal catch was reconstructed for 2011-2017 based on the number of artisanal vessels within marine zones (Table 1) and an approximate catch rate estimated from the total reconstructed artisanal catch-per-year for 2002-2010. The average reconstructed artisanal catch rate per vessel for 2006-2010 was estimated and applied to the number of vessels for 2011-2015 (Table 1). The total reconstructed artisanal catch was held constant for 2016-2017 because the number of vessels was not available for these years (Figure 1). Unreported artisanal catch was estimated for 2011-2017 as the remainder after subtracting the reported catches.

Table 1. Number of marine artisanal vessels and approximate catch rate per vessel (t·boat⁻¹·year⁻¹)

Year	Artisanal boats	Catch rate (t·boat ⁻¹ ·year ⁻¹)	Source
2002	464	23	DINARA (2008)
2003	486	30	DINARA (2008)
2004	519	46	DINARA (2008)
2005	531	50	DINARA (2008)
2006	572	41	DINARA (2008)
2007	566	26	DINARA (2008)
2008	480	43	DINARA (2009)
2009	394	79	DINARA (2010)
2010	491	72	DINARA (2012)
2011	522 ¹	52 ²	N. A.
2012	553	52 ²	DINARA (2013)
2013	469	52 ²	DINARA (2014a)
2014	494	52 ²	DINARA (2014b)
2015	466	52 ²	DINARA (2015)

¹Interpolated number of vessels between 2010 and 2012
²Average catch rate of 2006-2010 based on the reconstructed artisanal catch per vessel per year.

**Figure 1.** Total reconstructed domestic catch for Uruguay by fishing sector (1950-2018).

Transition from 2017 to 2018

The catches reconstructed to 2017 for Uruguay were carried forward to 2018 using the semi-automated procedure outlined in Noël (2020), based on FAO data. Semi-automated reconstructed catch data will be later replaced by a more detailed, research-intensive update.

Marine biodiversity protection

Uruguay has agreed to protect its biological diversity through the international agreements of the Convention on Biological Diversity (Aichi), and it is also part of the international network of UNESCO Man and the Biosphere (Marine Conservation Institute 2020).

Uruguay has seven MPAs and one marine managed area. Together, these areas cover 1,622 km² (Marine Conservation Institute 2020), corresponding to 1.22% of the EEZ of 133,000 km² (Lorenzo *et al.* 2016). “The National System of Protected Areas (SNAP) in Uruguay has become a fundamental tool to reconcile environmental care - in particular the diversity of landscapes, ecosystems, species and cultural elements - with the economic and social development of the country. To generate opportunities for local communities and society as a whole through recreation, tourism, education, research and the development of productive activities compatible with conservation” (Marine Conservation Institute 2020).

The MPA of Laguna de Rocha (a Protected Landscape designated in 2010 with a total area of 342 km² and a reported marine area of 102 km²; Marine Conservation Institute 2020) is exploited by two fishing communities (Mellado *et al.* 2014). An assessment carried out in the MPA of Laguna de Rocha to evaluate its contribution to its induction into the National System of Protected Areas of Uruguay showed that environmental issues are still rarely considered by national and municipal agencies. “Despite the affinity of the social actors toward the protected area, the high number and diversity of interests may become problematic if certain requirements are not met. Working in interdisciplinary teams is an opportunity to change the levels of reflection although several issues remain to be addressed. An issue that generated disagreement within the interdisciplinary team was the economic analysis, because an open epistemological discussion exists on the implications of the concept of economic price [and value of natural resources]” (Rodríguez-Gallego *et al.* 2013).

Currently, the Laguna de Rocha MPA has no specific measures to regulate fishery activities (Vögler *et al.* 2020). The fishery-ecological assessment between three areas of the SWA [(Uruguayan Southwest Atlantic)] revealed high catches of threatened coastal sharks inside the MPA-RLPL [(Protected Landscape of Laguna de Rocha)], and also huge catch differences among fishing fleets (artisanal and industrial), which are affecting some overexploited fish species. [...] Despite endangered fish species knowingly being caught in the MPA-RLPL there are no specific measures to regulate fishing activities in the area. The management plan for the Uruguayan MPAs needs to urgently include conservation measures to reduce fishing pressure on threatened coastal sharks and [enforce them and monitor their effectiveness]” (Vögler *et al.* 2020).

Venezuela

The catches of Venezuela’s marine fisheries were reconstructed for the years 1950 to 2010 by Mendoza (2015, 2016), and the text below is to document, if briefly, how the initial reconstruction was updated to 2017, then carried forward to 2018.

Reporting baseline

Venezuelan national data were not publicly available at the time this update was completed, and reported landings available from FAO were used as the reported baseline for 2011-2017. The FAO reported catch was

compared to nationally reported catch and found to be within 10% of the national data for the years 1950 to 2010. Thus, the reported catches were not retroactively changed.

The original FAO names within the data reported by the FAO were compared with taxon names used for the national reported data in 2010 to maintain the taxonomic resolution between the two data sources. The ratios of each reported species allocated to each sector and EEZ by year in the reported data in 2010 were applied to the reported taxa in the 2011-2017 FAO data, after excluding the industrial catch of tunas and other large pelagics (see Coulter *et al.* 2020) and catch taken from foreign EEZs.

All reconstructed catch taken from outside of the home EEZ were labelled as industrial for 2011-2017. Discards were applied to catch outside of the home EEZ at the same ratio as described for 2010 (Mendoza 2015). Unreported landings and discards for the industrial sector within the home EEZ were assumed to be zero for 2011-2017, because the trawl fishery is no longer active. Discards are assumed unlikely due to the current demand for food (Associated Press 2016). Artisanal unreported catch was calculated by taxa based on the 2010 ratios described in the original catch reconstruction (Mendoza 2015).

Subsistence and recreational catch

Unreported subsistence catches were estimated from the take-home catch from artisanal fisheries and catches taken for subsistence purposes by the Warao peoples, an indigenous group. Take home catch was calculated at 5% of total reconstructed artisanal catch for 2011-2017. Subsistence catch by the Warao people was calculated at 3% of the reconstructed catch of the estuarine species caught in the eastern subregion of the EEZ.

There has been no recreational catch reported since 1996; in light of the current political crisis, it has been assumed to remain at zero for all years after 1996.

Politics vs. fisheries

Due to Venezuela's current political and economic climate, there is very little information available about Venezuelan fisheries and maritime activities in recent years. Investigative journalists reported high rates of piracy and food riots due to extreme poverty in the country (Franklin 2018). The Associated Press (2016) estimated that coastal tuna catches have declined to less than a third of the tuna catch in 2004. Franklin (2018) estimates that catch fell to 226, 600 tonnes in 2015, a 60% decline from 1997 according to the Foundation for Sustainable and Responsible Tuna Fisheries.

Many private companies have left Venezuela to fish in other countries rather than sell half their catch to the government for Venezuelan currency (Associated Press 2016). Many domestic fishing boats are being repurposed for parts and sold, left to rust, or stolen (Associated Press 2016). Fishing has become a dangerous activity due to the increasing levels of violence, and many fishers have turned to smuggling to trade for essential items such as food, diapers or medicine (Associated Press 2016; Franklin 2018). Because of the turmoil and lack of information in recent years, there is a high degree of uncertainty in this reconstruction for 2011-2017.

Transition from 2017 to 2018

The catches reconstructed to 2017, however uncertain, were carried forward to 2018 using the semi-automated procedure outlined in Noël (2020), based on FAO landing data. Semi-automated reconstructed catch data will be later replaced by a more detailed, research-intensive update.

Marine biodiversity protection

Venezuela has agreed to protect its biodiversity through the international agreements of the Convention on Biological Diversity (Aichi) and Ramsar Convention on Wetlands of International Importance (Marine Conservation Institute 2020).

Venezuela has 25 MPAs and six marine managed areas which jointly cover 7,925 km² (Marine Conservation Institute 2020), corresponding to 1.7% of the EEZ of 474,770 km² (Mendoza 2016). “Protected areas in Venezuela are also intrinsically related to the cultural heritage and diversity of indigenous groups. For the Ye'kwana and Yanomani, as well as for the Pemon and the Hosti groups, territories within protected areas constitute their livelihood spaces, including: hunting and gathering sites, sacred sites, settlements, oral traditions, customary laws and other social manifestations. [...] Tourism in Venezuela's national parks is also an important activity, mostly for local people that live within protected areas. One of the most popular parks visited over the last few years was Parque Nacional Morrocoy [(designated in 1974 with a total area of 320 km² and a reported marine area of 192 km²; Marine Conservation Institute 2020)], with an annual average of 1.5 million visitors (Cartaya 2007)” (Secretariat of the Convention on Biological Diversity 2008).

“Current conservation efforts in Venezuela are deficient given the collapse of government institutions responsible for the management of fisheries. This has led [for example] to noncompliance with fishing regulations designed for elasmobranchs” (Marquez 2019). It is currently unclear to what extent the political instability, humanitarian crisis and poverty that the Venezuelan people are experiencing is affecting artisanal fisheries inside and outside the country's MPAs and marine managed areas.

Ideally, greater effort shall be made to assess biodiversity and ultimately conserve ecosystems and sensitive species in Venezuela, such as green turtles, which are known to use the Gulf of Venezuela as a feeding ground (Barrios-Garrido *et al.* 2020). “However, future advances in scientific research and conservation will depend largely on the recovery of the economic and political stability in Venezuela” (Marquez 2019).

Discussion

The countries of South America share in common limitations of the data reported by these countries to the FAO, if they report at all (Garibaldi 2012). The FAO reported catch resolution in space and by sector was improved using national reported catch statistics and secondary sources. For some countries (for example, Brazil and Venezuela), national data were no longer available for the update years. Therefore, data released by the FAO were relied upon to update catch reconstructions. Reporting of catch statistics, including where catches were taken, sector and fishing gear, and species caught, continues to remain important for management of fisheries.

Acknowledgments

Funding for the work presented here was provided by the Minderoo Foundation, by the Marisla, Oak, and David and Lucille Packard Foundations, and by the Bloomberg Philanthropies through Rare.

References (General, then by country)

- De la Puente, S., S.-L. Noël and R. White. 2020. Peru: updated catch reconstruction for 1950-2018, p. 249-278. In: B. Derrick, M. Khalfallah, V. Relano, D. Zeller and D. Pauly (eds). *Updating to 2018 the 1950-2010 Marine Catch Reconstructions of the Sea Around Us. Part II: The Americas and Asia-Pacific*. Fisheries Centre Research Report 28(6).
- Garibaldi, L. 2012. The FAO global capture production database: A six-decade effort to catch the trend. *Marine Policy*, 36: 760-768.

Pauly, D., and T. Charles. 2015. Counting on small-scale fisheries. *Science*, 347: 242-243.

Argentina

- Becherucci, M.E., A.F. Rosenthal and J.P. Seco Pon. 2017. Marine debris in beaches of the Southwestern Atlantic: An assessment of their abundance and mass at different spatial scales in northern coastal Argentina. *Marine Pollution Bulletin*, 119(1): 299-306.
- Elías, I., C. Carozza, E.E. Di Giacomio, M.S. Isla, J.M. Orensanz, A.M. Parma, R.C. Pereiro, M.R. Perier, R.G. Perrotta, M.E. Ré and C. Ruarte C. 2011. Coastal fisheries of Argentina, p. 13-48. *In*: S. Salas, R. Cheunpagdee, A. Charles and J.C. Seijo (eds). *Coastal fisheries of Latin America and the Caribbean. FAO Fisheries and Aquaculture Technical Paper No. 544*. Food and Agriculture Organization of the United Nations (FAO), Rome.
- FAO. 2012. Recreational fisheries. FAO Technical Guidelines for Responsible Fisheries No.13. Food and Agriculture Organization of the United Nations (FAO), Rome.
- Marine Conservation Institute. 2020. MPAtlas [Online]. Seattle, WA. Available at: www.mpatlas.org
- Morea, J.P. 2019. A framework for improving the management of protected areas from a social perspective: The case of Bahía de San Antonio Protected Natural Area, Argentina. *Land Use Policy*, 87: 104044.
- Noël, S.-L. 2020. Semi-automation procedure for catch reconstruction forward carry, p. 15-20. *In*: B. Derrick, M. Khalfallah, V. Relano, D. Zeller and D. Pauly (eds). *Updating to 2018 the 1950-2010 Marine Catch Reconstructions of the Sea Around Us: Part I – Africa, Antarctica, Europe and the North Atlantic*. Fisheries Centre Research Report 28(5).
- Venerus, L.A. and P.V. Cedrola. 2017. Review of marine recreational fisheries regulations in Argentina. *Marine Policy*, 81: 202-210.
- Villasante, S., G. Macho, J. Isusu de Rivero, E. Divovich, K. Zylich, S. Harper, D. Zeller and D. Pauly. 2015. Reconstruction of Argentina's marine fisheries catches (1950-2010). Fisheries Centre Working Paper #2015-50, 6 p.
- Villasante, S., G. Macho, J. Isusu de Rivero, E. Divovich, K. Zylich, S. Harper, D. Zeller and D. Pauly. 2016. Argentina, p. 190. *In*: D. Pauly and D. Zeller (eds). *Global Atlas of Marine Fisheries: A critical appraisal of catches and ecosystem impacts*. Island Press, Washington, D.C.
- Venerus, L.A. and P.V. Cedrola. 2017. Review of marine recreational fisheries regulations in Argentina. *Marine Policy*, 81: 202-210.

Brazil (mainland)

- Anon. 2017. Brasil consome, por ano, 14 kg de peixe por pessoa. *EBC Rádios*, 10 April, 2017. Available at: radios.ebc.com.br/cotidiano/2017/04/brasil-consome-anualmente-14-kg-de-peixe-por-pessoa
- Freire, K.M.F., J.A.N. Aragão, A.R.R. Araújo, A.O. Ávila-da-Silva, M.C.S. Bispo, G. Velasco, M.H. Carneiro, F.D.S. Gonçalves, K.A. Keunecke, J.T. Mendonça, P.S. Moro, F.S. Motta, G. Olavo, P.R. Pezzuto, R.F. Santana, R.A. Santos, I. Trindade-Santos, J.A. Vasconcelos, M. Vianna and E. Divovich. 2015. Reconstruction of catch statistics for Brazilian marine waters (1950-2010, p. 3-30. *In*: K.M.F. Freire and D. Pauly (eds). *Fisheries catch reconstructions for Brazil's mainland and oceanic islands*. Fisheries Centre research report 23(4).
- Freire, K.M.F., J.A.N. Aragão, A.R.R. Araújo, A.O. Ávila-da-Silva, M.C.S. Bispo, G. Velasco, M.H. Carneiro, F.D.S. Gonçalves, K.A. Keunecke, J.T. Mendonça, P.S. Moro, F.S. Motta, G. Olavo, P.R. Pezzuto, R.F. Santana, R.A. Santos, I. Trindade-Santos, J.A. Vasconcelos, M. Vianna and E. Divovich. 2016. Brazil, p. 206. *In*: D. Pauly and D. Zeller (eds). *Global Atlas of Marine Fisheries: A critical appraisal of catches and ecosystem impacts*. Island Press, Washington, D.C.
- Freire, K.M.F., D. Belhabib, J.C. Espedido, L. Hood, K.M. Kleisner, V.W.L. Lam, M.L. Machado, J.T. Mendonça, J.J. Meeuwig, P.S. Moro, F.S. Motta, M.L.D. Palomares, N. Smith, L. Teh, D. Zeller, K. Zylich and D. Pauly. 2020. Estimating global catches of marine recreational fisheries. *Frontiers in Marine Sciences*, 7(12). doi.org/10.3389/fmars.2020.00012
- Freire, K.M.F., U. Sumaila, D. Pauly and G. Adelino. 2018. The offshore recreational fisheries of northeastern Brazil. *Latin American Journal of Aquatic Research*, 46(4): 765-778.
- Giraldi-Costa, A.C., R.P. Medeiros and L.M. Tiepolo. 2020. Step zero of marine protected areas of Brazil. *Marine Policy*, 120: 104119.
- Marine Conservation Institute. 2020. MPAtlas [Online]. Seattle, WA. Available at: www.mpatlas.org

- Noël, S.-L. 2020. Semi-automation procedure for catch reconstruction forward carry, p. 15-20. In: B. Derrick, M. Khalfallah, V. Relano, D. Zeller and D. Pauly (eds). *Updating to 2018 the 1950-2010 Marine Catch Reconstructions of the Sea Around Us: Part I – Africa, Antarctica, Europe and the North Atlantic*. Fisheries Centre Research Report 28(5).
- Rhormens, M.S., A.D.G. Pedrini and N.P. Ghilardi-Lopes. 2017. Implementation feasibility of a marine ecotourism product on the reef environments of the marine protected areas of Tinharé and Boipeba Islands (Cairu, Bahia, Brazil). *Ocean and Coastal Management*, 139: 1-11.
- SECAP/AVALIACAO. 2019. Caderno de avaliação de gastos diretos. Avaliação Executiva do Seguro Defeso - Edição 1. Pátria Amada Brasil. Available at: antigo.economia.gov.br/Economia/central-de-conteudos/publicacoes/notas-informativas/2019/caderno_segurodefeso.pdf/view
- Sganzerla, T. 2017. Are Brazil's fisheries heading towards collapse? Without official data, no one knows. *OCEANA*, 3 February, 2017. Available at: oceana.org/blog/are-brazil%E2%80%99s-fisheries-heading-towards-collapse-without-official-data-no-one-knows

Brazil (Oceanic islands)

- Agardy, T., G.N. di Sciara and P. Christie. 2011. Mind the gap: Addressing the shortcomings of marine protected areas through large scale marine spatial planning. *Marine policy*, 35(2): 226-232.
- Anon. 2002. Trindade 2002. PWØT DXpedition. Trindade History. Available at: www.k7bv.com//Trindade/History/
- Anon. 2018. Brazil: Fernando de Noronha. Instituto Brasileiro de Geografia e Estatística. Available at: www.citypopulation.de/en/brazil/cities/fernandodenoronha/
- Divovich, E. and D. Pauly. 2016a. Brazil (Fernando de Noronha), p. 207. In: D. Pauly and D. Zeller (eds). *Global Atlas of Marine Fisheries: A critical appraisal of catches and ecosystem impacts*. Island Press, Washington, D.C.
- Divovich, E. and D. Pauly. 2016b. Brazil (Trindade & Martim Vaz Islands), p. 209. In: D. Pauly and D. Zeller (eds). *Global Atlas of Marine Fisheries: A critical appraisal of catches and ecosystem impacts*. Island Press, Washington, D.C.
- Divovich, E. and D. Pauly. 2015. Oceanic island of Brazil: catch reconstruction from 1950 to 2010, p. 31-48. In: K.M.F. Freire and D. Pauly (eds). *Fisheries catch reconstructions for Brazil's mainland and oceanic islands*. Fisheries Centre research report 23(4).
- Fellet, J. 2018. Governo excluiu aos '45 do 2º tempo' áreas mais vulneráveis de reservas marinhas, dizem pesquisadores. *BBC News Brasil*, 22 March 2018. Available at: www.bbc.com/portuguese/brasil-43498210
- Freire, K.M.F., U. Sumaila, D. Pauly and G. Adelino. 2018. The offshore recreational fisheries of northeastern Brazil. *Latin American Journal of Aquatic Research*, 46(4): 765-778.
- Lopes, P. and S. Villasante. 2018. Paying the price to solve fisheries conflicts in Brazil's Marine Protected Areas. *Marine Policy*, 93: 1-8.
- Marine Conservation Institute. 2020. MPAtlas [Online]. Seattle, WA. Available at: www.mpatlas.org
- Pauly, D., J.A. Vasconcelos and E. Divovich. 2016. Brazil (St. Peter and St. Paul Archipelago), p. 208. In: D. Pauly and D. Zeller (eds). *Global Atlas of Marine Fisheries: A critical appraisal of catches and ecosystem impacts*. Island Press, Washington, D.C.
- Rodrigues, R. 2018. Pesca é regulamentada em áreas protegidas marinhas. Instituto Chico Mendes de Conservação da Biodiversidade (ICMBio), Ministério do Meio Ambiente, Brazil. Available at: www.icmbio.gov.br/portal/ultimas-noticias/20-geral/9997-pesca-e-regulamentada-em-novas-areas-protegidas-marinhas
- Santos, A., W. Mohriak, M. Geraldies, W. Sanots, C. Ponte-Neto and N. Stanton. 2018. Compiled potential field data and seismic surveys across the Eastern Brazilian continental margin integrated with new magnetometric profiles and stratigraphic configuration for Trindade Island, South Atlantic, Brazil. *International Geology Review*, 61(14): 1728-1744.
- Schiavetti, A., J. Manz, C. Zapelini dos Santos, T.C. Magro and M.I. Pagani. 2013. Marine Protected Areas in Brazil: An ecological approach regarding the large marine ecosystems. *Ocean and Coastal Management*, 76: 96-104.
- Sganzerla, T. 2017. Are Brazil's fisheries heading towards collapse? Without official data, no one knows. *OCEANA*, 3 February, 2017. Available at: oceana.org/blog/are-brazil%E2%80%99s-fisheries-heading-towards-collapse-without-official-data-no-one-knows

Chile (mainland, and Juan Fernández/Desventuradas Islands)

- Alarcon, R., L. Cubillos and C. Gatica. 2008. Jumbo squid (*Dosidicus gigas*) biomass off central Chile: Effects on Chilean hake (*Merluccius gayi*). *CalCOFI Reports*. 49 p.
- Lee, J. 2015. Chile Creates Largest Marine Reserve in the Americas. *National Geographic News*, 15 October, 2015. Available at: www.nationalgeographic.com/news/2015/10/151005-desventuradas-islands-marine-protected-area-conservation-science/
- Marine Conservation Institute. 2020. MPAtlas [Online]. Seattle, WA. Available at: www.mpatlas.org
- Noël, S.-L. 2020. Semi-automation procedure for catch reconstruction forward carry, p. 15-20. In: B. Derrick, M. Khalfallah, V. Relano, D. Zeller and D. Pauly (eds). *Updating to 2018 the 1950-2010 Marine Catch Reconstructions of the Sea Around Us: Part I – Africa, Antarctica, Europe and the North Atlantic*. Fisheries Centre Research Report 28(5).
- Oyanedel, R., A. Marín, J.C. Castilla and S. Gelcich. 2016. Establishing marine protected areas through bottom-up processes: Insights from two contrasting initiatives in Chile. *Aquatic Conservation: Marine and Freshwater Ecosystems*, 26(1): 184-195.
- Petit, I. J., A.N. Campoy, M.J. Hevia, C.F. Gaymer and F.A. Squeo. 2018. Protected areas in Chile: are we managing them? *Revista chilena de historia natural*, 91(1): 1-8.
- SERNAPESCA. 2016. Anuarios Estadísticos. Servicio Nacional de Pesca y Acuicultura, Santiago.
- van der meer, L., H. Arancibia, K. Zylich and D. Zeller. 2015. Reconstruction of total marine fisheries catches for mainland Chile (1950-2010). Fisheries Centre Working Paper #2015-91, 16 p.
- van de Meer, L., H. Arancibia and D. Zeller. 2016. Chile, p. 218. In: D. Pauly and D. Zeller (eds). *Global Atlas of Marine Fisheries: A critical appraisal of catches and ecosystem impacts*. Island Press, Washington, D.C.
- Zylich, K. and L. van der Meer. 2015. Reconstruction of total marine fisheries catches for Juan Fernández Islands and the Desventuradas Islands (Chile) for 1950-2010. Fisheries Centre Working Paper #2015-92, 14 p.
- Zylich, K. and L. van der Meer. 2016. Chile (Juan Fernández and Desventuradas Islands), p. 220. In: D. Pauly and D. Zeller (eds). *Global Atlas of Marine Fisheries: A critical appraisal of catches and ecosystem impacts*. Island Press, Washington, D.C.

Chile (Easter Island)

- Aburto, J.A., C.F. Gaymer, S. Haoa and L. González. 2015. Management of marine resources through a local governance perspective: Re-implementation of traditions for marine resource recovery on Easter Island. *Ocean & Coastal Management*, 116: 108-115.
- Castilla, J.C., E. Yáñez, C. Silva and M. Fernández. 2014. A review and analysis of Easter Island's traditional and artisan fisheries. *Latin America Journal of Aquatic Resources*, 42(4): 690-702.
- Hernández, M., C. García-Moro, P. Moral and A. González-Martín. 2000. Population Evolution in 20th-century Easter Island: Endogamy and Admixture. *Human Biology*, 72(2): 359-377.
- Noël, S.-L. 2020. Semi-automation procedure for catch reconstruction forward carry, p. 15-20. In: B. Derrick, M. Khalfallah, V. Relano, D. Zeller and D. Pauly (eds). *Updating to 2018 the 1950-2010 Marine Catch Reconstructions of the Sea Around Us: Part I – Africa, Antarctica, Europe and the North Atlantic*. Fisheries Centre Research Report 28(5).
- SERNAPESCA. 2016. Estadísticas de desembarque por caletas. Gobierno de Chile. Servicio Nacional de Pesca (SEERNAPESCA). Available at: webmail.sernapesca.cl/Qry/Desembarque/artesanal.asp?
- Vaughan, A. 2015a. Chile plans world's biggest marine park to protect Easter Island fish stocks. *The Guardian*, 13 September, 2015. Available at: www.theguardian.com/environment/2015/sep/13/chile-plans-easter-island-marine-park-fishing
- Vaughan, A. 2015b. Chile to create one of world's largest marine parks around Easter Island. *The Guardian*, 5 October, 2015. Available at: www.theguardian.com/environment/2015/oct/05/chile-creates-one-of-worlds-largest-marine-parks-around-easter-island
- Zylich, K., S. Harper, R. Licandeo, R. Vega, D. Zeller and D. Pauly. 2014. Fishing in Easter Island, a recent history (1950-2010). *Latin American Journal of Aquatic Research*, 42(4): 845-856.
- Zylich, K., S. Harper, R. Licandeo, R. Vega, D. Zeller and D. Pauly. 2016. Chile (Easter Island), p. 219. In: D. Pauly and D. Zeller (eds). *Global Atlas of Marine Fisheries: A critical appraisal of catches and ecosystem impacts*. Island Press, Washington, D.C.

Colombia (Pacific and Caribbean)

- Coulter, A., T. Cashion, A. Cisneros-Montemayor, S. Popov, G. Tsui, F. Le Manach, L. Schiller, M. Palomares, D. Zeller and D. Pauly. 2020. Using harmonized historical catch data to infer the expansion of global tuna fisheries. *Fisheries Research*, 221:105379. doi.org/10.1016/j.fishres.2019.105379
- Cuervo-Sánchez, R., J.H. Maldonado and M. Rueda. 2018. Spillover from marine protected areas on the pacific coast in Colombia: A bioeconomic modelling approach for shrimp fisheries. *Marine Policy*, 88: 182-188.
- Digital Observatory for Protected Areas. 2019. Colombia. Available at: <https://dopa-explorer.jrc.ec.europa.eu/country/CO>
- Lindop, A.M.M., T. Chen, K. Zylich and D. Zeller. 2015. A Reconstruction of Colombia's marine fisheries catches. Fisheries Centre Working Paper #2015-32, 16 p.
- Marine Conservation Institute. 2020. MPAtlas [Online]. Seattle, WA. Available at: www.mpatlas.org
- Noël, S.-L. 2020. Semi-automation procedure for catch reconstruction forward carry, p. 15-20. In: B. Derrick, M. Khalfallah, V. Relano, D. Zeller and D. Pauly (eds). *Updating to 2018 the 1950-2010 Marine Catch Reconstructions of the Sea Around Us: Part I – Africa, Antarctica, Europe and the North Atlantic*. Fisheries Centre Research Report 28(5).
- Ramirez, L.F. 2016. Marine protected areas in Colombia: Advances in conservation and barriers for effective governance. *Ocean & coastal management*, 125: 49-62.
- Wielgus, J., D. Zeller, D. Caicedo-Herrera and R. Sumaila. 2010. Estimation of fisheries removals and primary economic impact of the small-scale and industrial marine fisheries in Colombia. *Marine Policy*, 34(3): 506-513.
- Wielgus, J., D. Caicedo-Herrera and D. Zeller. 2007. Reconstruction of Colombia's marine fisheries catches, p. 69-79. In: D. Zeller and D. Pauly (eds). *Reconstruction of Marine Fisheries Catches for Key Countries and Regions (1950-2005)*. Fisheries Centre research report 15(2).
- Wielgus, J., D. Caicedo-Herrera, A. Lindop, T. Chen, K. Zylich and D. Zeller. 2016a. Columbia (Caribbean), p. 223. In: D. Pauly and D. Zeller (eds). *Global Atlas of Marine Fisheries: A critical appraisal of catches and ecosystem impacts*. Island Press, Washington D.C
- Wielgus, J., D. Caicedo-Herrera, A. Lindop, T. Chen, K. Zylich and D. Zeller. 2016b. Columbia (Pacific), p. 224. In: D. Pauly and D. Zeller (eds). *Global Atlas of Marine Fisheries: A critical appraisal of catches and ecosystem impacts*. Island Press, Washington, D.C.

Ecuador (mainland)

- Alava, J.J., A. Lindop and J. Jacquet. 2015. Marine Fisheries Catch Reconstructions for Continental Ecuador: 1950-2010. Fisheries Centre Working Paper #2015-34, 25 p.
- Alava, J.J., A. Lindop and J. Jacquet. 2016. Ecuador, p. 241. In: D. Pauly and D. Zeller (eds). *Global Atlas of Marine Fisheries: A critical appraisal of catches and ecosystem impacts*. Island Press, Washington, D.C.
- Brotz, L., A. Schiariti, J. López-Martínez, J. Álvarez-Tello, Y.H.P. Hsieh, R.P. Jones, J. Quiñones, Z. Dong, A.C. Morandini, M. Preciado and E. Laaz. 2017. Jellyfish fisheries in the Americas: origin, state of the art, and perspectives on new fishing grounds. *Reviews in Fish Biology and Fisheries*, 27(1): 1-29.
- CIESIN. 2012. National Aggregates of Geospatial Data Collection: Population, Landscape, And Climate Estimates, Version 3 (PLACE III). Center for International Earth Science Information Network (CIESIN), Columbia University, NASA Socioeconomic Data and Applications Center (SEDAC), Palisades, NY. Available at: sedac.ciesin.columbia.edu/data/set/nagdc-population-landscape-climate-estimates-v3
- Coulter, A., T. Cashion, A. Cisneros-Montemayor, S. Popov, G. Tsui, F. Le Manach, L. Schiller, M. Palomares, D. Zeller and D. Pauly. 2020. Using harmonized historical catch data to infer the expansion of global tuna fisheries. *Fisheries Research*, 221:105379. doi.org/10.1016/j.fishres.2019.105379
- Gravez, V., R. Bensted-Smith, P. Heylings and T. Gregoire-Wright. 2013. Governance Systems for Marine Protected Areas in Ecuador, p. 145-158. In: E. Moksness, E. Dahl and J. Støttrup (eds). *Global Challenges in Integrated Coastal Zone Management*. John Wiley & Sons, Ltd, Oxford, UK.
- Jacquet, J., J.J. Alava, G. Pramod, S. Henderson and D. Zeller. 2008. In hot soup: sharks captured in Ecuador's waters. *Environmental Sciences*, 5(4): 269-283.
- Marine Conservation Institute. 2020. MPAtlas [Online]. Seattle, WA. Available at: www.mpatlas.org

- Noël, S.-L. 2020. Semi-automation procedure for catch reconstruction forward carry, p. 15-20. In: B. Derrick, M. Khalfallah, V. Relano, D. Zeller and D. Pauly (eds). *Updating to 2018 the 1950-2010 Marine Catch Reconstructions of the Sea Around Us: Part I – Africa, Antarctica, Europe and the North Atlantic*. Fisheries Centre Research Report 28(5).
- World Bank. 2018. World Population. Available at: data.worldbank.org/indicator/sp.pop.totl
- Quezada-Sarmiento, P.A., J.D.C. Macas-Romero, C. Roman and J.C. Martin. 2018. A body of knowledge representation model of ecotourism products in southeastern Ecuador. *Heliyon*, 4(12): e01063.
- Zambrano-Monserrate, M.A., C.A. Silva-Zambrano and M.A. Ruano. 2018. The economic value of natural protected areas in Ecuador: A case of Villamil Beach National Recreation Area. *Ocean & coastal management*, 157: 193-202.

Ecuador (Galapagos)

- Barragan-Paladines, M.J. and R. Chuenpagdee. 2017. A Step Zero Analysis of the Galapagos Marine Reserve. *Coastal Management*, 45(5): 339-359.
- Burbano, D.V., T.C. Meredith and M.E. Mulrennan. 2020. Exclusionary decision-making processes in marine governance: The rezoning plan for the protected areas of the 'iconic' Galapagos Islands, Ecuador. *Ocean & Coastal Management*, 185: 105066.
- Castrejón, M. and A. Charles. 2020. Human and climatic drivers affect spatial fishing patterns in a multiple-use marine protected area: The Galapagos Marine Reserve. *PLoS ONE*, 15(1): e0228094.
- Cerutti-Pereyra, F., N. Moity, M. Dureuil, J. Ramírez-González, H. Reyes, K. Budd, J.M. Jarrín and P. Salinas-de-León. 2019. Artisanal longline fishing the Galapagos Islands –effects on vulnerable megafauna in a UNESCO World Heritage site. *Ocean and Coastal Management*, 183: 104995.
- Coulter, A., T. Cashion, A. Cisneros-Montemayor, S. Popov, G. Tsui, F. Le Manach, L. Schiller, M. Palomares, D. Zeller and D. Pauly. 2020. Using harmonized historical catch data to infer the expansion of global tuna fisheries. *Fisheries Research*, 221:105379. doi.org/10.1016/j.fishres.2019.105379
- Denkinger, J. and L. Vinuela. 2014. *The Galapagos Marine Reserve: a dynamic social-ecological system*. Springer International Publishing, New York. 314 p.
- Eddy, T.D., A.M. Friedlander and P. Salinas de León P. 2019. Ecosystem effects of fishing & El Niño at the Galápagos Marine Reserve. *PeerJ*, 7: e6878.
- Giraldi-Costa, A.C., R.P. Medeiros and L.M. Tiepolo. 2020. Step zero of marine protected areas of Brazil. *Marine policy*, 120: 104119.
- Moity, N. 2018. Evaluation of no-take zones in the Galápagos marine reserve, zoning plan 2000. *Frontiers in Marine Science*, 5(244).
- Noël, S.-L. 2020. Semi-automation procedure for catch reconstruction forward carry, p. 15-20. In: B. Derrick, M. Khalfallah, V. Relano, D. Zeller and D. Pauly (eds). *Updating to 2018 the 1950-2010 Marine Catch Reconstructions of the Sea Around Us: Part I – Africa, Antarctica, Europe and the North Atlantic*. Fisheries Centre Research Report 28(5).
- Ramírez, J., H. Reyes and A. Schuhbauer. 2013. Evaluation of the spiny lobster fishery in the Galapagos Marine Reserve, p. 149-155. In: GNPS, GCREG, CDF and GC. *Galapagos Report 2011-2012*. Puerto Ayora, Galapagos, Ecuador.
- Reyes, H., J. Ramírez and A. Schuhbauer. 2013. Evaluation of the sea cucumber fishery in the Galapagos Marine Reserve, p. 143-148. In: GNPS, GCREG, CDF and GC. *Galapagos Report 2011-2012*. Puerto Ayora, Galapagos, Ecuador.
- Schiller, L., J.-J. Alava, J. Grove, G. Reck and D. Pauly. 2013. A reconstruction of fisheries catches for the Galápagos islands, 1950-2010. Fisheries Centre Working Paper #2013-11, 37 p.
- Schiller, L., J.-J. Alava, J. Grove, G. Reck and D. Pauly. 2015. The demise of Darwin's fishes: evidence of fishing down and illegal shark finning in the Galápagos Islands. *Aquatic Conservation: Marine and Freshwater Ecosystems*, 25(3): 431-446.
- Schiller, L., J.-J. Alava, J. Grove, G. Reck and D. Pauly. 2016. Ecuador (Galapagos), p. 242. In: D. Pauly and D. Zeller (eds). *Global Atlas of Marine Fisheries: A critical appraisal of catches and ecosystem impacts*. Island Press, Washington, D.C.

French Guiana

- Castelblanco-Martínez, D.N., V. dos Reis and B. de Thoisy. 2018. How to detect an elusive aquatic mammal in complex environments? A study of the Endangered Antillean manatee *Trichechus manatus manatus* in French Guiana. *Oryx*, 52(2): 382-392.

- Chambault, P., L. Giraudou, B. de Thoisy, M. Bonola, L. Kelle, V. D. Reis, F. Blanchard, Y. Le Maho and D. Chevallier. 2017. "Habitat use and diving behaviour of gravid olive ridley sea turtles under riverine conditions in French Guiana". *Journal of marine systems*, 165: 115-123.
- Cissé, A.A., F. Blanchard and O. Guyader. 2014. Sustainability of tropical small-scale fisheries: Integrated assessment in French Guiana. *Marine Policy*, 44: 397-405.
- Cormier-Salem, M.C. 2017. Mangrove grabbing: an exploration of changes in mangrove tenure from a political ecology perspective, p. 143-162. In: H. Artaud and A. Surallés (eds). *The sea within: maritime tenure and cosmopolitical debates*. IWGIA, Copenhagen, Denmark.
- Geographic Guianas. n.d. Marine biodiversity in French Guiana: Looking to the future by Marion Brichet (Agence des Aires Marines Protégées). Available at: www.guianas-geographic.com/article-en/biodiversity/marine-biodiversity-in-french-guiana-looking-to-the-future/#:~:text=French%20Guiana%20is%20also%20playing,and%20the%20Amana%20Nature%20Reserves
- Harper, S., L. Frotté, S. Booth, L. Veitch and D. Zeller. 2016. France (Guiana), p. 258. In: D. Pauly and D. Zeller (eds). *Global Atlas of Marine Fisheries: A critical appraisal of catches and ecosystem impacts*. Island Press, Washington, D.C.
- Harper, S., L. Frotté, S. Booth, L. Veitch and D. Zeller. 2015. Reconstruction of marine fisheries catches for French Guiana from 1950-2010. Fisheries Centre Working Paper #2015-07, 11 p.
- Marine Conservation Institute. 2020. MPAtlas [Online]. Seattle, WA. Available at: www.mpatlas.org
- Martinez, L., D. Gerales, A. Suardi, V. Wyss, E. Dutrieux and C. Chaineau. 2019. New sightings records of marine mammals and seabirds off French Guiana. *Latin American Journal of Aquatic Research*, 47(5): 753.
- Noël, S.-L. 2020. Semi-automation procedure for catch reconstruction forward carry, p. 15-20. In: B. Derrick, M. Khalfallah, V. Relano, D. Zeller and D. Pauly (eds). *Updating to 2018 the 1950-2010 Marine Catch Reconstructions of the Sea Around Us: Part I – Africa, Antarctica, Europe and the North Atlantic*. Fisheries Centre Research Report 28(5).
- UNEP-WCMC and IUCN. 2020. Protected Planet: Protected Area Profile for French Guyana from the World Database of Protected Areas, June 2020. Available at: www.protectedplanet.net/country/GF

Guyana

- Anon. 2015. Guyana fishermen could be allowed to bear arms. *Jamaica Observer*, 27 June, 2015. Available at: www.jamaicaobserver.com/news/Guyana-fishermen-could-be-allowed-to-bear-arms_19152807
- Anon. 2016a. Ramjattan proposes arming fisherman against pirates. *Stabroek News*, 6 June, 2016. Available at: www.stabroeknews.com/2016/06/06/news/guyana/ramjattan-proposes-arming-fishermen-pirates/
- Anon. 2016b. UN agency announces world's first illegal fishing treaty now in force. *UN News Centre*, 5 June, 2016. Available at: www.un.org/apps/news/story.asp?NewsID=54140#.V1myMrsrLIU
- Elias-Roberts, A. 2020. Balancing Environmental Protection and Offshore Petroleum Developments in Guyana. *Global Energy Law and Sustainability*, 1(1): 1-27.
- FAO. 2016. Agreement on Port State Measures to Prevent, Deter and Eliminate Illegal, Unreported and Unregulated Fishing. Food and Agriculture Organization of the United Nations (FAO), Rome. 115 p.
- Howard, J., M.A. Trotz, K. Thomas, E. Omisca, H.T. Chiu, T. Halfhide, F. Akiwumi, R. Michael and A.L. Stuart. 2011. Total mercury loadings in sediment from gold mining and conservation areas in Guyana. *Environmental monitoring and assessment*, 179(1-4): 555-573.
- MacDonald, J., S. Harper, S. Booth and D. Zeller. 2015. Guyana fisheries catch: 1950-2010. Fisheries Centre Working Paper #2015-21, 18 p.
- MacDonald, J., S. Harper, S. Booth, D. Zeller and D. Pauly. 2016. Guyana, p. 286. In: D. Pauly and D. Zeller (eds). *Global Atlas of Marine Fisheries: A critical appraisal of catches and ecosystem impacts*. Island Press, Washington, D.C.
- Marine Conservation Institute. 2020. MPAtlas [Online]. Seattle, WA. Available at: www.mpatlas.org
- Noël, S.-L. 2020. Semi-automation procedure for catch reconstruction forward carry, p. 15-20. In: B. Derrick, M. Khalfallah, V. Relano, D. Zeller and D. Pauly (eds). *Updating to 2018 the 1950-2010 Marine Catch Reconstructions of the Sea Around Us: Part I – Africa, Antarctica, Europe and the North Atlantic*. Fisheries Centre Research Report 28(5).
- Shapovalova, D. 2020. Indigenous Rights and Resource Extraction in Guyana: A Learning Opportunity for the New Offshore Sector? *Oil, Gas and Energy Law (OGEL)*, 18(2). doi.org/10.2139/ssrn.3555121

Xinhua. 2016. Fishermen attacked by pirates near Guyana, 1 killed, 3 missing. *Xinhua News*, 29 May, 2016. Available at: news.xinhuanet.com/english/2016-05/29/c_135396357.htm

Suriname

- Coulter, A., T. Cashion, A. Cisneros-Montemayor, S. Popov, G. Tsui, F. Le Manach, L. Schiller, M. Palomares, D. Zeller and D. Pauly. 2020. Using harmonized historical catch data to infer the expansion of global tuna fisheries. *Fisheries Research*, 221:105379. doi.org/10.1016/j.fishres.2019.105379
- Djosetro, M. and J.H. Behagel. 2020. Building local support for a coastal protected area: Collaborative governance in the Bigi Pan Multiple Use Management Area of Suriname. *Marine Policy*, 112: 103746.
- Hornby, C., S. Harper, J. MacDonald and D. Zeller. 2016. Suriname, p. 403. In: D. Pauly and D. Zeller (eds). *Global Atlas of Marine Fisheries: A critical appraisal of catches and ecosystem impacts*. Island Press, Washington, D.C.
- Hornby, C., S. Harper, J. MacDonald and D. Zeller. 2015. Reconstruction of Suriname's marine fisheries catches from 1950-2010. Fisheries Centre Working Paper #2015-49, 14 p.
- Marine Conservation Institute. 2020. MPAtlas [Online]. Seattle, WA. Available at: www.mpatlas.org
- Noël, S.-L. 2020. Semi-automation procedure for catch reconstruction forward carry, p. 15-20. In: B. Derrick, M. Khalfallah, V. Relano, D. Zeller and D. Pauly (eds). *Updating to 2018 the 1950-2010 Marine Catch Reconstructions of the Sea Around Us: Part I – Africa, Antarctica, Europe and the North Atlantic*. Fisheries Centre Research Report 28(5).

Uruguay

- Defeo, O., S. Horta, A. Carranza, D. Lercari, A. Álava, J. Gómez, G. Martínez, J. Lozoya and E. Celentano. 2009. Hacia un manejo ecosistémico de pesquerías. Áreas marinas protegidas en Uruguay. Facultad de Ciencias-DINARA, Montevideo. 122 p.
- DINARA. 2008. Boletín Estadístico Pesquero (2002-2007). Dirección Nacional de Recursos Acuáticos (DINARA). Ministerio de Ganadería, Agricultura y Pesca. 80 p.
- DINARA. 2009. Boletín Estadístico Pesquero 2008. Dirección Nacional de Recursos Acuáticos (DINARA). Ministerio de Ganadería, Agricultura y Pesca. 48 p.
- DINARA. 2010. Boletín Estadístico Pesquero 2009. Dirección Nacional de Recursos Acuáticos (DINARA). Ministerio de Ganadería, Agricultura y Pesca. 52 p.
- DINARA. 2012. Boletín Estadístico Pesquero 2010. Dirección Nacional de Recursos Acuáticos (DINARA). Ministerio de Ganadería, Agricultura y Pesca. 63 p.
- DINARA. 2013. Boletín Estadístico Pesquero 2012. Dirección Nacional de Recursos Acuáticos (DINARA). Ministerio de Ganadería, Agricultura y Pesca. Montevideo, 83 p.
- DINARA. 2014a. Boletín Estadístico Pesquero 2013. Dirección Nacional de Recursos Acuáticos (DINARA). Ministerio de Ganadería, Agricultura y Pesca. 73 p.
- DINARA. 2014b. Boletín Estadístico Pesquero 2014. Dirección Nacional de Recursos Acuáticos (DINARA). Ministerio de Ganadería, Agricultura y Pesca. 51 p.
- DINARA. 2015. Boletín Estadístico Pesquero 2015. Dirección Nacional de Recursos Acuáticos (DINARA). Ministerio de Ganadería, Agricultura y Pesca. 60 p.
- Gianelli, I., S. Horta, G. Martínez, A. de la Rosa, O. Defeo. 2018. Operationalizing an ecosystem approach to small-scale fisheries in developing countries: The case of Uruguay. *Marine Policy*, 95: 180-188.
- Ligrone, A., V. Franco-Trecu, C. Passadore, M. Szephegyi and A. Carranza. 2014. Fishing strategies and spatial dynamics of artisanal fisheries in the Uruguayan Atlantic coast. *Latin America Journal of Aquatic Resources*, 42(5): 1126-1135.
- Lorenzo, M.I., O. Defeo, N. Roshan Moniri and K. Zylich. 2015. Fisheries catch statistics for Uruguay. Fisheries Centre Working Paper #2015-25, 6 p.
- Lorenzo, M.I., O. Defeo, N. Roshan Moniri and K. Zylich. 2016. Uruguay, p. 439. In: D. Pauly and D. Zeller (eds). *Global Atlas of Marine Fisheries: A critical appraisal of catches and ecosystem impacts*. Island Press, Washington, D.C.
- Marine Conservation Institute. 2020. MPAtlas [Online]. Seattle, WA. Available at: www.mpatlas.org
- Mellado, T., T. Brochier, J. Timor and J. Vitancurt. 2014. Use of local knowledge in marine protected area management. *Marine Policy*, 44, 390-396. doi.org/10.1016/j.marpol.2013.10.004
- Noël, S.-L. 2020. Semi-automation procedure for catch reconstruction forward carry, p. 15-20. In: B. Derrick, M. Khalfallah, V. Relano, D. Zeller and D. Pauly (eds). *Updating to 2018 the 1950-2010 Marine Catch Reconstructions of the Sea Around Us: Part I – Africa, Antarctica, Europe and the North Atlantic*. Fisheries Centre Research Report 28(5).

- Rodríguez-Gallego, L., C. Santos, S. Amado, D. Gorfinkiel, M.N. González, J. Gómez, C. Neme, H. Tommasino and D. Conde. 2013. Interdisciplinary diagnosis and scenario analysis for the implementation of a coastal protected area, Laguna de Rocha Uruguay, p.389-411. In: A. Yáñez-Arancibia, R. Dávalos-Sotelo, J.W. Day and E. Reyes (eds). *Ecological Dimensions for Sustainable Socio-Economic Development*. WIT Press, Southampton, Boston.
- Vögler, R., C. González and A.M. Segura. 2020. Spatio-temporal dynamics of the fish community associated with artisanal fisheries activities within a key marine protected area of the southwest Atlantic (Uruguay). *Ocean and Coastal Management*, 190: 105175. doi.org/10.1016/j.ocecoaman.2020.105175

Venezuela

- Associated Press. 2016. Venezuela: Pirates Terrorizing Fishermen as Industry Crumbles. *NBC News*, 8 December, 2016. Available at: www.nbcnews.com/news/latino/venezuela-pirates-terrorizing-fishermen-industry-crumbles-n693586
- Barrios-Garrido, H., P. Becker, K.A. Bjorndal, A.B. Bolten, C.E. Diez, N. Espinoza-Rodríguez, M. Fastigi, J. Gray, E. Harrison, K.A. Hart, A. Meylan, P. Meylan, M.G. Montiel-Villalobos, F. Morales, M. Nava, J. Palmar, M.J. Petit-Rodriguez, P. Richardson, K.M. Rodríguez-Clark, D. Rojas-Cañizales, M.G. Sandoval, R.A. Valverde, R. van Dam, J.T. Walker, N. Wildermann and M. Hamann. 2020. Sources and movements of marine turtles in the Gulf of Venezuela: Regional and local assessments. *Regional studies in marine science*, 36: 101318.
- Cartaya, V. 2007. Conservación y Bienestar Humano en Venezuela: El Aporte de las Áreas Protegidas. Síntesis del Informe Final para la Fundación. The Nature Conservancy of Venezuela. 75 p.
- Coulter, A., T. Cashion, A. Cisneros-Montemayor, S. Popov, G. Tsui, F. Le Manach, L. Schiller, M. Palomares, D. Zeller and D. Pauly. 2020. Using harmonized historical catch data to infer the expansion of global tuna fisheries. *Fisheries Research*, 221:105379. doi.org/10.1016/j.fishres.2019.105379
- Franklin, J. 2018. Venezuelan Pirates Rule the Most Lawless Market on Earth. *Bloomberg Businessweek*, 30 January, 2018. Available at: www.bloomberg.com/news/features/2018-01-30/venezuelan-pirates-rule-the-most-lawless-market-on-earth
- Marine Conservation Institute. 2020. MPAtlas [Online]. Seattle, WA. Available at: www.mpatlas.org
- Marquez, R. 2019. Elasmobranch species in the artisanal fishery of Sucre State, Venezuela. *Ciencias marinas*, 45(4): 181-188.
- Mendoza, J. 2015. Rise and fall of Venezuelan industrial and artisanal marine fisheries: 1950-2010. Fisheries Centre Working Paper #2015-27, 15 p.
- Mendoza, J. 2016. Venezuela, p. 454. In: D. Pauly and D. Zeller (eds). *Global Atlas of Marine Fisheries: A critical appraisal of catches and ecosystem impacts*. Island Press, Washington, D.C.
- Noël, S.-L. 2020. Semi-automation procedure for catch reconstruction forward carry, p. 15-20. In: B. Derrick, M. Khalfallah, V. Relano, D. Zeller and D. Pauly (eds). *Updating to 2018 the 1950-2010 Marine Catch Reconstructions of the Sea Around Us: Part I – Africa, Antarctica, Europe and the North Atlantic*. Fisheries Centre Research Report 28(5).
- Secretariat of the Convention on Biological Diversity. 2008. Protected Areas in Today's World: Their Values and Benefits for the Welfare of the Planet. Technical Series no. 36. Secretariat of the Convention on Biological Diversity, Montreal. vii + 96 pages.